Spectral Sequences of Flow Categories

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Abstract

Framed flow categories and their stable homotopy types are central tools in modern Floer, contact, Khovanov, and knot homotopy theories. In this paper, given an $\mathcal{X}$-structured flow category $\mathscr{C}$ and a generalized homology theory A equipped with a map $M\mathcal{X} \to A$, we construct an Atiyah-Hirzebruch-type spectral sequence which converges to the homology of the realization $A_*(|\mathscr{C}|)$. Moreover, if A is a bordism theory - for example, if $\mathscr{C}$ is framed and A is stable homotopy - we provide a description of the higher pages and differentials in terms of gluing the flow manifolds in $\mathscr{C}$ to a geometric analog of Toda brackets, an idea traced back to Floer[11].

This work provides a bordism-level perspective on spectral sequences, aligning with the approach of Abouzaid and Blumberg, who developed a bordism model for spectra. We demonstrate these spectral sequence techniques by calculating the unoriented and oriented bordism homology of real projective spaces using Morse flow data.

Independent of flow categories, we provide a description of the spectral sequence of a chain complex in a stable category in terms of morphisms between chain complexes.

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1 Introduction

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Short Introduction

Flow categories originally emerged as a categorical bookkeeping mechanism to organize the geometric and analytic data extracted from Hamiltonian systems on a manifold [8]. In recent years, flow categories and their stable homotopy types have become central tools for lifting invariants in Floer, contact, and Khovanov homology into the realm of stable homotopy theory. A natural question arises: how can we systematically compute the generalized homology of these realization spectra directly from the underlying geometric flow data?

Here is an intuitive description of the main theorem of the paper using a toy model coming from Morse theory. Let M be a closed manifold and $f : M \to \mathbb{R}$ a Smale Morse function. Denote by $\mathrm{Crit}(f)$ the set of critical points of f, and by $\mathrm{ind} : \mathrm{Crit}(f) \to \mathbb{Z}$ the index function. For simplicity, assume that there is one critical point for each index $i \in \{0, \ldots, n\}$, and denote the critical points by x_i where $\mathrm{ind}(x_i) = i$. The first result is:

There is a spectral sequence

$$E_1^{n,p} = \Omega^{\mathrm{fr}}_{n-p} \implies \pi_n \Sigma^\infty_+ M.$$

where Ω^{fr}_* is the framed bordism ring, and $\Sigma^\infty_+ M$ is the stable homotopy type of M.

Let x_j, x_i be two critical points of f with $j > i$. We denote by $\mathcal{M}(x_j, x_i)$ the compactification of the moduli space of flow lines from x_j to x_i, up to reparametrization. Recall that $\mathcal{M}(x_j, x_i)$ is a framed manifold with corners of dimension $\mathrm{ind}(x_j) - \mathrm{ind}(x_i) - 1$, and that the boundary structure is given by

$$\partial \mathcal{M}(x_j, x_i) = \bigcup_{j>r>i} \mathcal{M}(x_j, x_r) \times \mathcal{M}(x_r, x_i).$$

Let W_p be a representative of a bordism class in $E_1^{n,p} = \Omega^{\mathrm{fr}}_{n-p}$. In our framework, we view W_p as a right flow module of degree n over the trivial subcategory containing only the critical point x_p. The main theorem states that W_p survives to the $(r+1)$-th page (also denoted by $W_p \in Z_{r+1}$) if and only if it **extends to a right flow module** W of degree n over the truncated flow category $\mathscr{C}_{[p,p-r]}$. Concretely, this means providing a family of framed manifolds $\{W_s\}$ for $s \in \{p, \ldots, p-r\}$ of dimension $n - s$ such that the corner structure maps

$$W_t \times \mathcal{M}(x_t, x_s) \hookrightarrow W_s$$

satisfy the axioms of a right flow module.

To compute the differential, we glue this right module W together with the canonical left flow module $\mathcal{M}(-, x_{p-(r+1)})$. Let $t, u \in \mathbb{Z}$ be such that $p \geq t > u \geq p - (r+1)$. By the boundary structure axioms of the right module W and the left module $\mathcal{M}(-, x_{p-(r+1)})$, we have the following two embeddings:

$$W_t \times \mathcal{M}(x_t, x_{p-(r+1)}) \hookleftarrow W_t \times \mathcal{M}(x_t, x_u) \times \mathcal{M}(x_u, x_{p-(r+1)}) \hookrightarrow W_u \times \mathcal{M}(x_u, x_{p-(r+1)})$$

and these boundary components are shared only by these two manifolds of the collection $W_s \times \mathcal{M}(x_s, x_{p-(r+1)})$. Because all boundary components of $W_s \times \mathcal{M}(x_s, x_{p-(r+1)})$ are of this form, we can glue the collection together along these shared faces to obtain a framed closed manifold. We denote this closed manifold by $W \circ \mathcal{M}(-, x_{p-(r+1)})$, and call it a Toda manifold:

$$W \circ \mathcal{M}(-, x_{p-(r+1)}) \coloneqq \bigcup_{s \in \{p,\dots,p-r\}} W_s \times \mathcal{M}(x_s, x_{p-(r+1)}).$$

This gluing operation is a geometric realization of the composition of modules (sometimes called a geometric tensor product).

To verify that this manifold represents a class in the correct group for the differential, we compute its dimension. For each s, the pieces being glued have dimension:

$$\dim(W_s \times \mathcal{M}(x_s, x_{p-(r+1)})) = \dim(W_s) + \dim(\mathcal{M}(x_s, x_{p-(r+1)})) = (n-s) + \big(s-(p-(r+1))-1\big) = n-p+r.$$

Note that the -1 in the second term arises from the dimension formula for flow manifolds. The result $n-p+r$ matches the expected dimension for an element of the $(r+1)$-th page. The last part of the theorem states that if W_p survives to Z_{r+1}, then the $r+1$ differential is given by the bordism class of this Toda manifold:

$$d_{r+1}(W_p) = [W \circ \mathcal{M}(-, x_{p-(r+1)})] \in \Omega^{\mathrm{fr}}_{n-p+r} = E_{r+1}^{n-1,p-(r+1)}.$$

See Fig. 2 for an example of the structure of the right module W and the flow manifolds, Fig. 3 for the gluing diagram of the Toda manifold, and Fig. 4 for the resulting closed manifold.

Background

Morse theory basics. The field of differential topology studies the connection between analysis on a smooth n-manifold M and its underlying topology. This field was pioneered by Morse in the beginning of the 20th century, and further developed by Smale, Thom, Milnor and more (see [20] for Morse first work and [18] for Milnor's classic exposition of the subject). To begin understanding M, we take a "nice" (namely a Smale-Morse) function $f\colon M \to \mathbb{R}$. We extract the *critical points* of f and, for any critical point p, assign its *index* $\mathrm{ind}(p)$ - which is defined as the signature of the Hessian of $-\nabla f$. Pictorially, this index represents the number of 'downward' axes at p, see Fig. 1.

For any two critical points y and x, we define $M(y,x)$ to be the moduli space of curves (up to reparametrization) in M that start at y and end at x, and follow the gradient field $-\nabla f$. We call such curves *flow lines*; see Fig. 1. Under generic conditions, $M(y,x)$ is a smooth (not necessarily closed) manifold of dimension $\mathrm{ind}(y) - \mathrm{ind}(x) - 1$.

Framed flow manifolds. Our main interests are *flow manifolds*, which are the compactification

$$\mathcal{M}(y,x) \coloneqq \overline{M(y,x)}.$$

The compactification is given by adding *broken flow lines*, which are piecewise smooth curves that start at y and end at x and pass through other critical points. We get that the flow manifolds have a *corner structure*

$$\partial\mathcal{M}(z,x) = \bigcup_{\mathrm{ind}(z)>\mathrm{ind}(y)>\mathrm{ind}(x)} \mathcal{M}(z,y) \times \mathcal{M}(y,x).$$

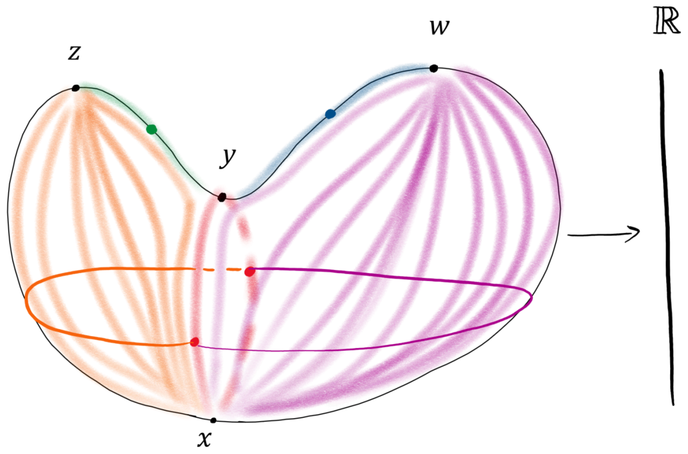

Figure 1: A Smale-Morse function on the sphere, given as projection to the z axis of the heart-shaped embedding. The letters represent critical points (with indices 2,2,1,0), the flow lines are drawn as soft, colorful lines, the solid colorful dots are representatives for the zero-dimensional flow manifolds and the solid colorful lines are representatives for the one-dimensional flow manifolds (notice the boundary structure as broken flow lines)

see Fig. 1. The fact that $\mathcal{M}(y, x)$ is constructed form the transverse intersection of two discs (one for curves ascending to y and one for curves descending to x) gives a stable *framing* on $\mathcal{M}(y, x)$ - a trivialization of the form

$$T\mathcal{M}(y, x) \oplus \underline{\mathbb{R}}^{\ell_1} \simeq \underline{\mathbb{R}}^{\ell_2}$$

where $\underline{\mathbb{R}}^{\ell_i}$ are trivial vector bundles over $\mathcal{M}(y, x)$. Moreover, the framing of all the flow manifolds is compatible with the corners structure.

Morse homology. It's worth saying that this data is sufficient to calculate the homology $H_*(M, \mathbb{Z})$. It is given as the homology of the Morse chain complex (C_*, d). The generators of C_p are critical points with index p. Given two critical points y, x with $\mathrm{ind}(y) - 1 = \mathrm{ind}(x)$, the flow manifold $\mathcal{M}(y, x)$ is in fact compact; the coefficient of x in $d(y)$ is given by the signed count of $\mathcal{M}(y, x)$, where the signs are determined by the framing. The boundary structure of the one-dimensional moduli spaces $\mathcal{M}(z, x)$ shows that $d^2 = 0$. See [24].

Bordism and generalized homology. The fact that the flow manifolds are framed hints at an important theorem in differential topology, the Pontryagin-Thom isomorphism. We say that two closed framed n-manifolds M_1 and M_2 are *bordant* if there is a framed $(n+1)$-manifold W, with $\partial W = M_1 \sqcup M_2$, in a way that is compatible with the framing. We denote by Ω_n^{fr} the set of n dimensional framed manifolds up to bordism. In fact the collection Ω_*^{fr} form a graded ring, by disjoint union and Cartesian product.

The *Pontryagin-Thom isomorphism* states that there is an isomorphism

$$\Omega_n^{\mathrm{fr}} \simeq \operatorname*{colim}_k [S^{n+k}, S^k]$$

between bordism groups of framed manifolds and stable homotopy groups of spheres (see [21, 25]). The right hand side is also known as the coefficient ring $\pi_n(\mathbb{S})$, where $\mathbb{S}$ is the spectrum representing stable homotopy. We can view the Morse chain complex as a chain complex with coefficient ring $\mathbb{Z} = \Omega_0^{\mathrm{fr}} = \mathbb{S}_{\leq 0}$ and the differential is given by multiplication by the 0 dimensional flow manifolds $\mathcal{M}(y, x)$.

Floer theory A related question is the analysis of real-valued functions on objects that are not finite-dimensional closed manifolds. Floer developed this idea in the 1980s, leading to what is now known as Floer theory. In general, Floer's idea is to generate a chain complex in which the differentials are defined by 'counting solutions' to a moduli problem (see [10] for initial formulations of this idea). In this case, the outcome does not yield the homology of a manifold, and one may ask: *Are Floer homology groups merely the shadow of a more structured object?*

Flow categories. The development of a framework to address these questions led Cohen, Jones, and Segal (CJS) in [8] to introduce *framed flow categories*, which axiomatize the data extracted from a Smale-Morse function on a manifold. In short, a framed flow category $\mathscr{C}$ consists of a set of objects (corresponding to critical points) with a grading function $|-|$ (corresponding to the index). For two objects $y, x \in \mathscr{C}$, the morphism space from y to x is an $(|y| - |x| - 1)$-dimensional manifold with corners, denoted $\mathcal{M}_{y,x}$, equipped with a choice of trivialization of $T\mathcal{M}_{y,x} \oplus \underline{\mathbb{R}^k}$ for some k (corresponding to the stable framing). Moreover, the composition is given by identifying the faces in the following way:

$$\mathcal{M}_{z,y} \times \mathcal{M}_{y,x} \hookrightarrow \mathcal{M}_{z,x}.$$

See Section 3.3 for definitions and details. Another building block in the theory are *flow modules*. Given a flow category $\mathscr{C}$, a *right flow module* W over $\mathscr{C}$ (see Definition 3.24 for the formal definition) consists of a collection of manifolds $\{W_x\}_{x\in\mathscr{C}}$, together with corner structure maps

$$W_y \times \mathcal{M}_{y,x} \hookrightarrow W_x.$$

A right flow module is of *degree* i if

$$\dim W_x = i - |x|$$

for each pair of objects y and x in $\mathscr{C}$. See [22] definition 4.6, 4.7. By abuse of notation, we denote the collection of right modules of degree i over $\mathscr{C}$

$$[*[i], \mathscr{C}].$$

Similarly, we can define left flow modules of degree i over $\mathscr{C}$ (see Definition 3.25 for the formal definition) to be collections of manifolds $\{V_x\}_{x\in\mathscr{C}}$, together with corner structure maps

$$\mathcal{M}_{y,x} \times V_x \hookrightarrow V_y.$$

such that $\dim V_x = |x| - i$. We denote the collection of left modules of degree i over $\mathscr{C}$ by

$$[\mathscr{C}, *[i]].$$

A useful example for left module on a category $\mathscr{C}$ is given by fixing $x' \in \mathscr{C}$ and taking the collection

$$\{\mathcal{M}_{x,x'}\}_{x\in\mathscr{C}}. \tag{1}$$

Indeed we have that the corner structure is given by the maps

$$\mathcal{M}_{y,x} \times \mathcal{M}_{x,x'} \hookrightarrow \mathcal{M}_{y,x'}$$

The degree of this module is given by

$$|x| - \dim(V_x) = |x| - (|x| - |x'| - 1) = |x'| + 1$$

and we denote it simply by $\mathcal{M}_{-,x'}$.

Filtrations and coherent chain complexes. Another result of CJS was to attach a stable homotopy type to a framed flow category by the following idea.

Given a filtered space $X_0 \to X_1 \to \cdots \to X_n = X$ we can take the successive cofibers to produce a "chain complex" defined by

$$K_* \coloneqq X_n/X_{n-1} \xrightarrow{d} \Sigma(X_{n-1}/X_{n-2}) \xrightarrow{d} \ldots \xrightarrow{d} \Sigma^n X_0,$$

where d^2 is null-homotopic via the canonical null homotopy arising from X. Moreover, viewing K_* from higher-categorical perspective allows us to encode additional data. In particular, there exist two null homotopies for d^3 that can be concatenated to form a loop in a suitable mapping space; the chain complex is then equipped with a null homotopy of this loop, and so forth. We call this structure a *coherent chain complex*. We can attempt to recover X by taking *iterated cofibers*. For example, note that

$$\Sigma^2 X_1 = \operatorname{cof}\left(\Sigma^1 X_1/X_0 \to \Sigma^2 X_0\right), \quad \Sigma^2 X_2 = \operatorname{cof}\left(\operatorname{cof}(X_2/X_1 \to \Sigma X_1/X_0) \to \Sigma^2 X_0\right)$$

and indeed, we can recover X up to a certain suspension. See Section 2.1 and [4] for full detailed text.

Building on this idea, CJS showed that a framed flow category $\mathscr{C}$ can be collapsed,via a Pontryagin-Thom-type construction, into a coherent chain complex K in Spectra. After taking iterated cofiber, we obtain a spectrum

$$|\mathscr{C}| \coloneqq \lim_p \operatorname{cof}(\operatorname{cof}(\ldots \operatorname{cof}(\operatorname{cof}(K_p \to K_{p-1}) \to K_{p-2}) \to \cdots \to K_1) \to K_0)$$

which is called the *realization* of $\mathscr{C}$.

Remark. Note that $|\mathscr{C}|$ depends on the choice of framing for the manifolds $M_{b,a}$.

Example - Morse flow category. To combine the ideas of framed flow categories and chain complexes, let $f : M \to \mathbb{R}$ be a manifold equipped with a Smale-Morse function. First, we construct from f a framed flow category $\mathscr{C}_f$ - by taking $\mathcal{M}_{y,x} \coloneqq \mathcal{M}(y,x)$, then we collapse it to a chain complex $K_{\mathscr{C}_f}$ and realize it as $|\mathscr{C}_f| \in \mathbf{Sp}$. A result by CJS [8], states that this procedure recovers the stable homotopy type of M:

$$|\mathscr{C}_f| \simeq \Sigma^\infty_+ M.$$

This framework has proven useful in Floer theory, where the solutions to the moduli problem sometimes exhibit the structure of a flow category $\mathscr{C}$. In such cases, we can realize the flow category, demonstrating that Floer homology is indeed a shadow of $|\mathscr{C}|$.

Looking back to Morse homology, we observe that the homology of a flow category can be determined from its zero-dimensional morphism manifolds. The fact that we can recover the stable homotopy type of M from its framed flow category motivates the idea that we can read off the (generalized) homology of M using the flow manifolds.

This idea goes back to Floer, who pointed out in [11] that: If h_* is a general homology theory, trajectory space analysis suggests contributions from compact components of $\mathcal{M}(x,y)$.

For stable homotopy $h_* = \pi_*^s$, this corresponds to the framed cobordism type. While higher-dimensional components of $\mathcal{M}(x,y)$ may not be compact, a modified approach could yield a spectral sequence. This framework has limited use in finite-dimensional Morse theory but may apply to infinite-dimensional settings.

Notations

We denote by **Top** the 1-category of topological spaces, by **Spaces** the category of homotopy types (or ∞-groupoids), and **Sp** is the category of spectra. The stabilization functor is denoted $\Sigma_+^\infty : \mathbf{Spaces} \to \mathbf{Sp}$, and $\mathrm{Ch}(\mathbf{Sp})$ and $\widehat{\mathrm{Fil}}(\mathbf{Sp})$ denote the categories of coherent chain complexes in Spectra and (complete) filtered Spectra, respectively.

For a topological space Y, we denote by Y_+ its one-point compactification (if Y is compact, then $Y_+ = Y \cup \{*\}$). For a CW complex Y, we denote by $\mathrm{sk}_n Y$ its n-th skeleton.

For a stable tangential structure $\xi : \mathcal{X} \to BO$, we denote by $\Omega_*^{\mathcal{X}}$ the graded $\mathcal{X}$-bordism ring and by $M\mathcal{X}$ the Thom spectrum associated with the $\mathcal{X}$-structure (see Section 7 for definitions).

In this paper, a manifold is assumed to be a *closed manifold* unless stated otherwise.

Let $\mathscr{C}$ be a flow category. For $q, p \in \mathbb{Z}$ we denote by $\mathscr{C}_{[q,p]}$ the full subcategory of $\mathscr{C}$ consisting of objects satisfying $q \geq |x| \geq p$, and we define $\mathscr{C}_p := \mathscr{C}_{[p,p]}$. For brevity, we write

$$M_{p,q} := \bigsqcup_{y \in \mathscr{C}_q, x \in \mathscr{C}_p} M_{y,x}.$$

for the union of the flow manifolds, and for a right / left module over $\mathscr{C}$ we write

$$W_p := \bigsqcup_{x \in \mathscr{C}_p} W_x, \quad V_p := \bigsqcup_{x \in \mathscr{C}_p} V_x.$$

Moreover, for a stable tangential structure $\xi : \mathcal{X} \to BO$, one can define $\mathcal{X}$ flow categories as those flow categories in which the morphism manifolds are equipped with an $\mathcal{X}$ structure that is compatible with composition (in place of a stable framing). For an $\mathcal{X}$ flow category $\mathscr{C}$, we obtain a coherent chain complex $K_{\mathscr{C}} \in \mathrm{Ch}(\mathrm{Mod}_{M\mathcal{X}})$, and, analogously to the realization in **Sp** (see realization), we define $|\mathscr{C}| \in \mathrm{Mod}_{\mathcal{X}}$ to be its iterated cofiber.

Our results

Let $\mathcal{X}$ be a stable tangential structure, let $\mathscr{C}$ be a $\mathcal{X}$-structured flow category. An element $W_p = \{W_z\}_{|z|=p} \in E_1^{n,p}$ survives to the $(r+1)$-th page if and only if it extends to a right flow module W of degree n over $\mathscr{C}_{[p,p-r]}$ (see Definition 3.28 for the formal definition).

To describe the differential, let W, V be a right and left flow module over $\mathscr{C}$ of degree n and $p-r$ respectively (see Definition 3.24, Definition 3.25). We define the punctured cube diagram

$$D : \ P(\{p, \dots, p-r\}) \setminus \{\varnothing\} \to \mathbf{Top},$$

where $P(-)$ denote the power set, by setting

$$D(\{a_k, \dots, a_1\}) = W_{a_k} \times \mathcal{M}_{a_k, a_{k-1}} \times \mathcal{M}_{a_{k-1}, a_{k-2}} \cdots \times \mathcal{M}_{a_2, a_1} \times V_{a_1},$$

where $a_k > a_{k-1} > \cdots > a_1$. The **Toda manifold** is the colimit which we denote

$$V \circ W := \operatorname{colim} D$$

and we will see later that the Toda manifolds indeed inherits manifold structure from V, W, and that they are closed, $(n-p+r)$-dimensional, $\mathfrak{X}$-structured.

For the spectral sequence, we consider a family of such constructions indexed by objects in $\mathscr{C}_{p-(r+1)}$. Specifically, for each $z \in \mathscr{C}_{p-(r+1)}$, we define a punctured cube diagram D_z by replacing the left module V with $\mathcal{M}_{-,z}$ in the general construction. This yields a Toda manifold

$$\mathcal{M}_{-,z} \circ W := \operatorname{colim} D_z \in \Omega^{\mathfrak{X}}_{n-p+r}.$$

Taking the disjoint union over all $z \in \mathscr{C}_{p-(r+1)}$, we obtain an element

$$\bigsqcup_{z \in \mathscr{C}_{p-(r+1)}} (\mathcal{M}_{-,z} \circ W) \in \bigoplus_{z \in \mathscr{C}_{p-(r+1)}} \Omega^{\mathfrak{X}}_{n-p+r}.$$

Example. In Fig. 2, 3, and 4, we can see an example of an extension and its Toda manifold.

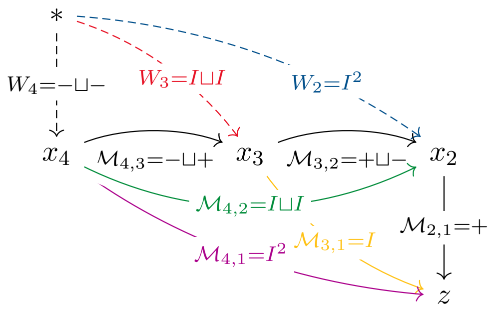

Figure 2: The solid part represents the flow category $\mathscr{C}$. Its objects are the x_i's with $|x_i| = i$, and the arrows represent the morphism manifolds. The 0 dimensional framed manifold W_4 represents an element in $E_1^{4,4} = \Omega^{\mathrm{fr}}_{4-4}$. The dashed arrows represent an right module over $\mathscr{C}_{[4,2]}$. The symbols $\pm$, I, etc., represent the flow manifolds themselves (where $\pm$ represents a framed point; the rest of framing is omitted for clarity).

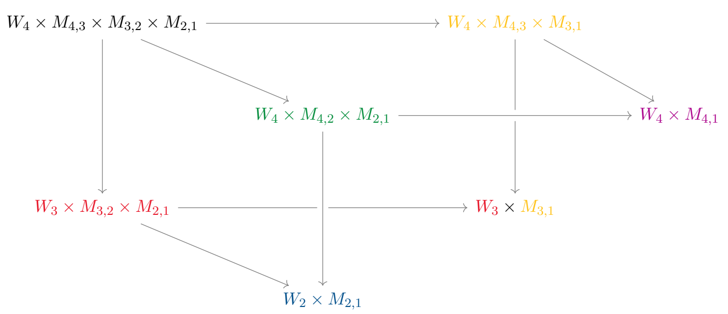

Figure 3: The punctured 3-cube diagram D_z associated with the object $z = x_1$ of degree 1, constructed from the right flow module W and the left flow module $\mathcal{M}_{-,z}$.

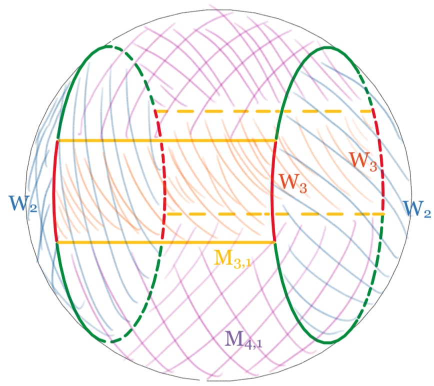

Figure 4: We can see that $\operatorname{colim} D_z$ is a topological space that inherits a closed smooth manifold structure.

Theorem A. *Let $\mathscr{C}$ be a flow category with an $\mathfrak{X}$-tangential structure. There is a spectral sequence:*

$$E_1^{n,p} = \bigoplus_{|z|=p} \Omega^{\mathfrak{X}}_{n-p}$$

which conditionally converges to the homotopy groups of the realization:

$$E_r^{n,p} \implies \pi_n \left|\mathscr{C}\right|.$$

An element

$$W_p = \{W_z\}_{|z|=p} \in E_1^{n,p}$$

survives to $E_{r+1}^{n,p}$ if and only if the collection W_p extends to a right flow module over $\mathscr{C}_{[p,p-r]}$ (see Definition 3.28). The differential is represented geometrically by the Toda manifolds:

$$d_{r+1}(W_p) = \bigcup_{y\in\mathscr{C}_{p-(r+1)}} (\mathcal{M}_{-,y}\circ W) \in \bigoplus_{y\in\mathscr{C}_{p-(r+1)}} \Omega^{\mathfrak{X}}_{n-p+r} = E_{r+1}^{n-1,p-(r+1)}$$

resolve: by Lurie we ahve that the S.S con-vernce to th homotopy groups of the colim of the filtered object. We never justi-fied why thi colim is $|\mathscr{C}|$

move this ref after the thm?

A key tool used to prove Theorem A is a description of the spectral sequence associated with a coherent chain complex expressed in terms of chain complex maps.

Theorem B. *Let $K \in \mathrm{Ch}(\mathbf{Sp})$ be a coherent chain complex in Spectra. Then there is a spectral sequence:*

$$E_1^{n,p} = \pi_{n-p}K_p,$$

which conditionally converges to

$$E_r^{n,p} \implies \pi_n \operatorname*{colim}_p (\mathcal{I}K)_p,$$

Moreover, let $x : \mathbb{S}^{n-p} \to K_p$ represent an element of $E_1^{n,p}$. Then x survives to $E_{r+1}^{n,p}$ if and only if there exists a chain complex map:

$$\begin{tikzcd}
\mathbb{S}^{n-p} \arrow[r] \arrow[d, "x"'] & 0 \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & 0 \arrow[d] \\
K_p \arrow[r, "d_p"] & K_{p-1} \arrow[r, "d_{p-1}"] & \dots \arrow[r, "d_{p-r}"] & K_{p-r}
\end{tikzcd} \tag{2}$$

Here, the top row is concentrated in degree p, and the bottom row is the restriction of K from degree p down to $p-r$. The differential $\partial_{r+1}(x)$ is given by the composition of chain maps:

$$\begin{tikzcd}
\mathbb{S}^{n-p} \arrow[r] \arrow[d, "x"'] & \dots \arrow[r] \arrow[d] & 0 \arrow[d] \\
K_p \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & K_{p-r} \arrow[d] \\
0 \arrow[r] & \dots \arrow[r] & K_{p-r-1}
\end{tikzcd}$$

The first map is precisely Eq. (2) and the second is induced by the structure of the chain complex K.

We obtain this result by combining the description of the spectral sequence of a filtered object in Lurie [15] with the results of [4].

Remark 1.1. If the flow category $\mathscr{C}$ (or the chain complex K) is bounded from below, the conditional convergence in Theorem A and Theorem B upgrades to strong convergence (referred to simply as convergence in [15, Prop. 1.2.2.14]). In general, the stable filtered object associated with a chain complex is complete, which allows us to use the Milnor exact sequence to satisfy Boardman's completeness criteria [6, Def. 5.10] and guarantee conditional convergence even when unbounded (see Section 2.2 for details).

Relation to other work

This paper contributes to the broader project of providing a geometric model for spectra (see [1]), building on the seminal work of Cohen, Jones, and Segal [8] who first realized Morse and Floer data as stable homotopy types.

In low-dimensional topology, flow categories are used to lift Khovanov homology to a stable homotopy type [12], where Pontryagin-Thom gluing of 1-dimensional moduli spaces computes the secondary cohomology operation Sq^2 [13]. Our work generalizes this gluing to higher-dimensional flow manifolds with corners to geometrically model all higher spectral sequence differentials and Toda brackets.

In symplectic geometry, flow categories and modules arise in Lagrangian Floer theory to construct a spectral Fukaya category over the sphere spectrum [22], where successive truncations of flow modules act as obstructions to lifting quasi-isomorphisms. A natural direction for future work is generalizing our spectral sequence framework to equivariant flow categories, such as the Morse-Bott models of Côté–Kartal [9] and the equivariant flow categories of Rezchikov [23] used to construct cyclotomic structures on Floer spectra.

Morphisms and invariance. In [1], the authors construct an ∞-category $\mathrm{Flow}^{\mathrm{fr}}$ (see also [22] for the homotopy category) and show that it is equivalent to $\mathbf{Sp}$. The morphisms in this category are *flow bimodules*. Given two Morse-Smale functions f and g on a compact manifold M, one obtains two framed flow categories, $\mathscr{C}_f$ and $\mathscr{C}_g$, and a flow bimodule between them, showing that the stable homotopy type is independent of the choice of Morse-Smale function. However, the different CW structures yield different spectral sequences, so the spectral sequence itself is not an invariant of the equivalence class in $\mathrm{Flow}^{\mathrm{fr}}$. To address this, we expect there is a construction analogous to that in [1] that defines another ∞-category with a modified class of morphisms. Specifically, for a 2-morphism R between flow categories $\mathscr{C}$ and $\mathscr{D}$, we require that if $\|y\| > \|x\|$, then $R_{x,y} = \varnothing$. This construction would provide a flow-categorical model for the subcategory of $\{K_p\} \in \mathrm{Ch}(\mathbf{Sp})$ satisfying $K_p = \bigoplus \mathbb{S}$. In this setup, the two Morse-Smale functions on a manifold M produce distinct objects, and we can think of Theorem A as a spectral sequence of a filtered flow category $F \in \widehat{\mathrm{Fil}}(\mathrm{Flow}^{\mathcal{X}})$ induced by the index filtration.

Remark 1.2. In this paper, we do not attempt to implement this full theoretical framework. The complete ∞-categorical structure is unnecessary for our purposes; instead, we only require a practical model for the composition of left and right flow modules to extract the geometric differentials of the spectral sequence.

The interpretation of gluing manifolds as Toda brackets or Massey products in cobordism appears in [2] (see also [19]), where the Massey product is defined via the gluing of manifolds with boundary.

Theorem B bears a similar flavor to other works on spectral sequences associated with various objects. In [4], the authors explain the connection between filtered objects and coherent chain complexes, and they also define a spectral sequence for coherent chain complexes using the Beilinson t-structure. We also recommend [3] as an introduction to spectral sequences and their relation both to chain complexes and to the spectral sequence of filtered objects defined in [15]. The construction of the spectral sequence in this paper is similar to that in [5], where the authors define a spectral sequence associated with a simplicial object.

Paper layout

Section 2 is devoted to the framework of chain complexes in (stable) ∞-categories and contains the proof of Theorem B.

Section 3 recalls the definitions of manifolds with corners, framed flow categories, and the Pontryagin-Thom construction for manifolds with corners. At the end of this section, we demonstrate several properties of the Pontryagin-Thom construction that will be useful in later sections.

Section 4 establishes a correspondence between chain complexes in the category of spectra (with values $\bigoplus \mathbb{S}$) and framed flow categories.

Our goal in Sections 5 and 6 is to upgrade this correspondence so that it interacts naturally with spectral sequences. In other words, given a framed flow category, we show how to produce a spectral sequence for the corresponding chain complex and provide a geometrical description of its pages and differentials.

In Section 5, we compare various models for chain complexes by considering different models for ∞-categories (namely, topologically enriched, simplicially enriched, and simplicial sets). This comparison allows us to give a straightforward description of the composition of maps between flow categories.

Section 6 translates this description of composition into a geometrical framework and establishes the existence of a spectral sequence associated with framed flow categories.

The final section explains how the framing structure can be replaced by a more general tangential structure, and discusses how different structures interact. This section also contains examples of spectral sequences arising from flow categories.

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2 Chains Complexes and their Spectral Sequences

Spectral sequences for filtered objects in Spectra are treated in Lurie's work [15]. A key insight, developed by Ariotta and Krause [4], is that within a stable ∞-category, filtered objects are equivalent to coherent chain complexes. Building upon this equivalence, this section aims to provide an explicit description of the spectral sequence associated with a coherent chain complex.

2.1 Chain Complexes and Filtered Objects

In this section, let $\mathcal{C}$ be a stable ∞-category with sequential limits and colimits. Following [4], we will introduce the categories of chain complexes and filtered objects in ∞-category. We will show how to connect between them and the equivalence between the two categories.

Remark 2.1. For being compatible with flow categories, our chain complexes will decrease and hence the filtered objects will increase, as oppose to [4].

Definition 2.2. We define the category Ch to be the pointed (ordinary) category with

$$\mathrm{ob}\ \ \mathrm{Ch} := \mathbb{Z} \cup \{*\}$$

$$\mathrm{Hom}_{\mathrm{Ch}}(i,j) := \begin{cases} \{\mathrm{id}, \mathbf{0}\} & i = j \\ \{d_i, \mathbf{0}\} & i = j+1 \\ \{\mathbf{0}\} & \text{otherwise} \end{cases}$$

where $*$ is the zero object and $\mathbf{0}$ is the map $i \to * \to j$.

For $\mathcal{C}$ as above, we define *chain complexes in* $\mathcal{C}$ (or just *chains*) to be the full subcategory

$$\mathrm{Ch}(\mathcal{C}) := \mathrm{Fun}^*(\mathrm{Ch}, \mathcal{C}) \subset \mathrm{Fun}(\mathrm{Ch}, \mathcal{C})$$

of pointed functors.

We denote by Ch_b the full subcategory of $\mathrm{Ch}(\mathcal{C})$ spanned by the objects K such that there exists $N \in \mathbb{N}$ with:

$$K_i = 0 \text{ for all } i < N.$$

We call such chain complexes *bounded below*.

Remark. If we don't mention otherwise, we assume that $K \in \mathrm{Ch}_b$ is bounded below by $N = 0$.

Example. The category $\mathrm{Ch}(\mathbf{Ab})$ of chain complexes in abelian groups is the usual category of chain complexes.

Remark 2.3. In the case where $\mathcal{C}$ is an ∞-category, the reader may want to think of $\mathrm{Ch}(\mathcal{C})$ as chain complexes, with null-homotopies of the composition of two maps along with higher null-homotopies: given a composition of three maps in the chain complex

$$K_p \to K_{p-1} \to K_{p-2} \to K_{p-3},$$

we get two null-homotopies of the map $K_p \to K_{p-3}$, which give us a map $S^1 \to \mathrm{Hom}(K_p, K_{p-3})$. Part of the data of a chain complex is a null-homotopy of this loop and similarly higher null-homotopies for higher loops.

Definition 2.4. We denote by $\mathbb{Z}$ the category of integers with the usual increasing order.

Definition 2.5. Let $\mathcal{C}$ be a stable ∞-category with sequential limits and colimits. We define the category of (increasing) complete *filtered objects in* $\mathcal{C}$ to be the full subcategory

$$\widehat{\mathrm{Fil}}(\mathcal{C}) \subset \mathrm{Fun}(\mathbb{Z}, \mathcal{C})$$

spanned by functors F such that $\lim_p F_p = 0$. We define the **bounded below filtered objects** to be the full subcategory

$$\widehat{\mathrm{Fil}}_{\mathrm{b}}(\mathcal{C}) \subset \widehat{\mathrm{Fil}}(\mathcal{C})$$

spanned by functors F such that there exists $N \in \mathbb{N}$ with $F_p = 0$ for $p < N$.

Note that the image of the stable Yoneda embedding

$$よ_{(-)} : \mathbb{Z}^{\mathrm{op}} \to \mathrm{Fun}(\mathbb{Z}, \mathbf{Sp}).$$

is given by

$$よ_p(q) = \mathrm{Map}_{\mathbf{Sp}}(p, q) = \begin{cases} \mathbb{S} & q \geq p \\ 0 & q < p \end{cases}$$

Let $p \to (p-1)$ be a morphism in $\mathbb{Z}^{\mathrm{op}}$, the morphisms $よ_p \to よ_{p-1}$ is given by

$$\begin{tikzcd} \dots \arrow[r] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \mathbb{S} \arrow[r] \arrow[d] & \dots \\ \dots \arrow[r] & 0 \arrow[r] & \mathbb{S} \arrow[r] & \mathbb{S} \arrow[r] & \dots \end{tikzcd}$$

Definition 2.6. For $n, p \in \mathbb{Z}$ we define the following chain complex $\mathbb{S}^n_p \in \mathrm{Ch}(\mathbf{Sp})$ by

$$\mathbb{S}^n_p(q) := \begin{cases} \mathbb{S}^n & q = p \\ 0 & \text{otherwise} \end{cases}$$

Note that there is the map in $\mathrm{Ch}(\mathbf{Sp})$

$$Y_{n,p} \colon \mathbb{S}^{n-1}_p \to \mathbb{S}^n_{p-1}$$

which visually given by

$$\begin{tikzcd} \dots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathbb{S}^{n-1} \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \dots \\ \dots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \mathbb{S}^n \arrow[r] & 0 \arrow[r] & \dots \end{tikzcd}$$

which is defined using the fact that $\mathbb{S}^n \simeq \Sigma \mathbb{S}^{n-1}$ in $\mathbf{Sp}$.

In particular we have a map

$$Y_p := Y_{-(p-1),p} \colon \mathbb{S}^{-p}_p \to \mathbb{S}^{-(p-1)}_{p-1}$$

Definition 2.7. We define
$$\mathcal{A}\colon \widehat{\mathrm{Fil}}(\mathbf{Sp}) \to \mathrm{Ch}(\mathbf{Sp})$$
to be the colimit preserving functor that determined by mapping
$$\mathcal{A}(\text{よ}_p) := \mathbb{S}_p^{-p}$$
and the morphisms are given by Y_p.

We denote it's right adjoint
$$\widehat{\mathrm{Fil}}(\mathbf{Sp}) \leftarrow \mathrm{Ch}(\mathbf{Sp})\colon \mathcal{I}$$

In [4], Ariotta define these functors and proves

Theorem 2.8 ([4])**.** *Let $\mathcal{C}$ be a stable ∞-category with sequential limits and colimits. We have an equivalence*
$$\mathcal{A} : \widehat{\mathrm{Fil}}(\mathcal{C}) \simeq \mathrm{Ch}(\mathcal{C}) : \mathcal{I}.$$

An immediate consequence is the representability of the evaluation:
$$\mathrm{Map}(\mathbb{S}_p^{-p}, \mathcal{A}F) \simeq \mathrm{Map}(\mathcal{I}\mathbb{S}_p^{-p}, F) \simeq \mathrm{Map}(\text{よ}_p, F) \simeq F_p, \tag{3}$$
which give us a formula for the p-th piece of $\mathcal{I}K$:
$$(\mathcal{I}K)_p \simeq \mathrm{Map}(\mathbb{S}_p^{-p}, K).$$

Definition 2.9. Let $\mathcal{C}$ be a stable ∞-category and $p \in \mathbb{Z}$, we write $\iota_{[p-1,p]} : \{(p-1) \to p\} \to \mathbb{Z}$ for the inclusion functor and
$$(\iota_{[p-1,p]})^* : \mathrm{Fun}(\mathbb{Z}, \mathcal{C}) \to \mathrm{Fun}(\Delta^1, \mathcal{C})$$
for the restriction (pre-composition) along $\iota_{[p-1,p]}$. We define the p-th associated graded functor by:
$$\mathrm{gr}_p := \mathrm{cof} \circ (\iota_{[p-1,p]})^* : \mathrm{Fun}(\mathbb{Z}, \mathcal{C}) \to \mathcal{C},$$
so that
$$F \mapsto F_p/F_{p-1}.$$
The map
$$F_p/F_{p-1} \to \Sigma F_{p-1}/F_{p-2}$$
given by the connecting morphism of the cofiber, and then taking quotient, induces a transformation:
$$\mathrm{gr}_p \to \Sigma\, \mathrm{gr}_{p-1}.$$

A direct calculation shows that
$$\Sigma^{-p} \mathrm{gr}_q(\text{よ}_p) = (\mathcal{A}\,\text{よ}_p)_q$$
and we deduce that

Corollary 2.10. FILLLL
$$(\mathcal{A}-)_p \simeq \Sigma^{-p} \mathrm{gr}_p(-)$$
As functors
$$\widehat{\mathrm{Fil}}(\mathbf{Sp}) \to \mathbf{Sp}$$

We now provide an explicit expression for $\mathcal{I}$.
First, we define a the following chain complex in $\mathbf{Sp}$,

$$\Sigma^n y_p := \begin{cases} \mathbb{S}^n & q \in \{p, p-1\} \\ 0 & \text{otherwise} \end{cases}$$

In particular we have

$$\mathrm{cof}(Y_p) \simeq \Sigma^{-(p-1)} y_p$$

On the other hand (see [4] Lemma 3.12), one can show that the evaluation functor

$$(-)_p \colon \mathrm{Ch}(\mathbf{Sp}) \to \mathbf{Sp}, \quad K \mapsto K_p$$

is represented:

$$\mathrm{Map}(y_p, -) \simeq (-)_p.$$

The objects in the following fiber sequence

$$\mathbb{S}_p^{-p} \to \mathbb{S}_{p-1}^{-(p-1)} \to \mathrm{cof}(Y_p) \tag{4}$$

represents the functors

$$(\mathcal{I}-)_p, \quad (\mathcal{I}-)_{p-1}, \quad \Sigma^{-(p-1)}(-)_p.$$

By taking the contravariant Map functor to Eq. (4) we get the following result:

Lemma 2.11. *There is a cofiber sequence of functors*

$$\Sigma^{-(p-1)}(-)_p \to (\mathcal{I}-)_{p-1} \to (\mathcal{I}-)_p \tag{5}$$

A simple corollary is

Corollary 2.12. *Let $K \in \mathrm{Ch}_b(\mathbf{Sp})$ bounded below by N. Then for $p \leq N$ we have*

$$(\mathcal{I}K)_p = 0$$

which obtained by inductively applying Lemma 2.11.

Example. Let

$$\cdots \to 0 \to F_0 \to F_1 \to F_2 \to F_3 \to 0 \to \cdots$$

be a filtered object in $\mathbf{Sp}$. Then $K := \mathcal{A}F$ is given by:

$$\cdots \to 0 \to \Sigma^{-3} F_3/F_2 \to \Sigma^{-2} F_2/F_1 \to \Sigma^{-1} F_1/F_0 \to F_0 \to 0 \cdots$$

where the zeros on the left are F_3/F_3. The composition

$$K_3 = \Sigma^{-3} F_3/F_2 \to \Sigma^{-2} F_2/F_1 \to \Sigma^{-1} F_1/F_0 = K_1$$

can be factored as:

$$\Sigma^{-3} F_3/F_2 \to \Sigma^{-2} F_2 \to \Sigma^{-2} F_2/F_1 \to \Sigma^{-1} F_1 \to \Sigma^{-1} F_1/F_0,$$

where the middle part is a cofiber sequence; hence, the composition is null-homotopic. Following Remark 2.3, this null-homotopy is the "non-higher" part of the chain complex K.

To recover $F = \mathcal{I}K$ from K, we can use Lemma 2.11. The fact that K is bounded below by -1 yields that

$$(\mathcal{I}K)_p = 0 \quad \text{for } p \leq -1$$

From that following cofiber sequence We ca see that $F_0 = K_0$

$$\Sigma^{-(-1)} K_0 \to \overbrace{F_{-1}}^{=0} \to F_0.$$

We can recover F_1 as the cofiber

$$K_1 \to \overbrace{F_0}^{=K_0} \to F_1.$$

And by taking F_2 is reconstructed as an iterated cofiber, from the cofiber sequence :

$$\Sigma^{-1} K_2 \to \overbrace{F_1}^{=K_1/K_0} \to F_2.$$

Remark 2.13. An "non example" is given by taking a topological space X and the filtered space $F_p := \operatorname{sk}_p X$. If X is finite, then there is an $N \in \mathbb{N}$ - an upper bound for the filtration. In this case, we can define the following chain complex in **Spaces**

$$K_p := \Sigma^{N-p} \operatorname{gr}_p F = \operatorname{sk}_p X / \operatorname{sk}_{p-1} X \simeq \bigvee_{\text{p cells}} S^N$$

and the morphisms $\operatorname{gr}_p \to \Sigma \operatorname{gr}_{p-1}$ that induces the maps $K_p \to K_{p-1}$, are given by the maps:

$$\bigvee S^p = \operatorname{sk}_p X / \operatorname{sk}_{p-1} X \xrightarrow{\partial_p} \operatorname{sk}_{p-1} X / \operatorname{sk}_{p-2} X = \Sigma \bigvee S^{p-1},$$

which are more commonly known as the differentials in the chain complex that computes the cellular homology of X. By taking iterated cofibers we can recover F only up to suspension, for example

$$\Sigma^N F_1 \simeq \operatorname{cof}(\Sigma^{N-1} F_1/F_0 \to S^N F_0)$$

2.2 Spectral Sequences of Chain Complexes

We will start with the construction of spectral sequence of a filtered object, following the notations and definitions in [16]. Combined with the last subsection we will get a description of the pages of a spectral sequence of a chain complex. The main output is Proposition 2.24.

For our discussion, we assume that $\mathcal{C}$ is a stable ∞ category equipped with a t-structure $(\mathcal{C}^{\leq 0}, \mathcal{C}^{\geq 0})$. We begin by defining the spectral sequence associated with a filtered object in $\mathcal{C}$.

Definition 2.14. [15, Def. 1.2.2.9] Let F be a filtered object in $\mathcal{C}$, i.e., a sequence of morphisms $\{F_p\}_{p \in \mathbb{Z}}$ such that

$$\cdots \to F_{p-1} \to F_p \to F_{p+1} \to \cdots . \tag{6}$$

For each $p \in \mathbb{Z}$, define the graded pieces by

$$\operatorname{gr}_p F = \operatorname{cof}(F_{p-1} \to F_p). \tag{7}$$

We define a spectral sequence $\{E_r^{n,p}, d_r^{n,p}\}$ associated with F by setting

$$E_r^{n,p} = \operatorname{Im}\left(\pi_n\left({F_p}/{F_{p-r}}\right) \to \pi_n\left({F_{p+r-1}}/{F_{p-1}}\right)\right), \tag{8}$$

where the map is induced by the compositions of the inclusion and projection:

$$F_p/F_{p-r} \to F_p/F_{p-1} \to F_{p+r-1}/F_{p-1}. \tag{9}$$

The differentials $d_r^{n,p} \colon E_r^{n,p} \to E_r^{n-1,p-r}$ are determined by the connecting morphisms via the commutativity of the diagram:

$$\begin{tikzcd}
\pi_n\left(F_p/F_{p-r}\right) \arrow[r] \arrow[d, "\delta_{p-r}"] & E_r^{n,p} \arrow[r] \arrow[d, "d_r^{n,p}"] & \pi_n\left(F_{p+r-1}/F_{p-1}\right) \arrow[d, "\delta_{p-1}"] \\
\pi_{n-1}\left(F_{p-r}/F_{p-2r}\right) \arrow[r] & E_r^{n-1,p-r} \arrow[r] & \pi_{n-1}\left(F_{p-1}/F_{p-r-1}\right),
\end{tikzcd} \tag{10}$$

where the vertical maps δ_{p-r} and δ_{p-1} are the connecting homomorphisms from the long exact sequences of homotopy groups associated with the cofiber sequences:

$$F_{p-r} \to F_p \to F_p/F_{p-r} \quad \text{and} \quad F_{p-1} \to F_{p+r-1} \to F_{p+r-1}/F_{p-1}. \tag{11}$$

Theorem 2.15. *[15, Prop. 1.2.2.14] Let $\mathcal{C}$ be a stable ∞-category equipped with a t-structure, that admits sequential colimits and that the t- structure is compatible with sequential colimits. Let F be a bounded below filtered object in $\mathcal{C}$ (see Definition 2.5). Then the spectral sequence $\{E_r^{n,p}, d_r^{n,p}\}$ (see Definition 2.14) converges to the homotopy groups of the colimit of F:*

$$E_r^{n,p} \Rightarrow \pi_n \operatorname*{colim}_p F_p.$$

Example 2.16. Consider a CW complex X with its skeletal filtration $\{\mathrm{sk}_p X\}_{p\geq 0}$. Applying the suspension spectrum functor Σ^∞, we obtain a filtered spectrum F in $\mathbf{Sp}$, where $F_p = \Sigma^\infty \mathrm{sk}_p X$. Let $K = \mathcal{A}F$ be the associated chain complex (as in Remark 2.13). The spectral sequence arising from this filtration has first page:

$$E_1^{n,p} = \pi_n \operatorname{gr}_p F = \pi_n\left(\bigoplus \mathbb{S}^p\right) = \pi_{n-p} K_p,$$

since each $\operatorname{gr}_p F$ is a wedge of spheres suspended p times, and K_p is the p-th term in the chain complex K.

This spectral sequence converges to:

$$\pi_n \Sigma^\infty X,$$

which are the stable homotopy groups of the space X.

More generally, for any spectrum A, we can consider the filtered spectrum $F_p \otimes A$, obtaining a spectral sequence converging to the generalized homology:

$$A_n(X).$$

This spectral sequence is known as the *Atiyah–Hirzebruch spectral sequence.*

Definition 2.17. Let F be a filtered object in stable ∞-category $\mathcal{C}$ and let $E_r^{n,p}$ be it's spectral sequence. An element $x \in E_1^{n,p}$ is said to **survive to the r-th page** if it lies in the image of the map:

$$x \in \operatorname{Im}\left(\pi_n\left({}^{F_p}/{}_{F_{p-r}}\right) \to \pi_n\left({}^{F_p}/{}_{F_{p-1}}\right)\right).$$

We denote the collection of such elements by $\mathcal{E}_r^{n,p}$. If $x' \in \pi_n\left({}^{F_p}/{}_{F_{p-r}}\right)$ maps to x, we say that x' **lifts** x.

Intuitively, an element survives to the r-th page if it is not killed by the differentials d_k for $1 \leq k < r$.

Remark 2.18. In the stable homotopy category **Sp**, the homotopy groups π_n are representable by spheres $\mathbb{S}^n$. Thus, elements in

$$E_1^{n,p} = \pi_n \operatorname{gr}_p F = \pi_{n-p} K_p$$

are represented by maps $x\colon \mathbb{S}^{n-p} \to K_p$.

We introduce a notation for subcategories of chain complexes.

Definition 2.19. For integers $a \leq b$, let $\mathrm{Ch}_{[b,a]}$ denote the full subcategory of the poset category Ch spanned by the objects

$$\{i \in \mathbb{Z} \mid b \geq i \geq a\} \cup \{*\}, \tag{12}$$

where $*$ denotes the initial object. Given a pointed category $\mathcal{C}$, we define

$$\mathrm{Ch}_{[b,a]}(\mathcal{C}) = \mathrm{Fun}^*(\mathrm{Ch}_{[b,a]}, \mathcal{C}) \subseteq \mathrm{Fun}(\mathrm{Ch}_{[b,a]}, \mathcal{C}), \tag{13}$$

the full subcategory of pointed functors from $\mathrm{Ch}_{[b,a]}$ to $\mathcal{C}$.

Precomposition with the inclusion $\mathrm{Ch}_{[b,a]} \hookrightarrow \mathrm{Ch}$ induces restriction functors:

$$i_{[b,a]}^*\colon\ \mathrm{Ch}(\mathcal{C}) \to \mathrm{Ch}_{[b,a]}(\mathcal{C}).$$

Given $K \in \mathrm{Ch}(\mathcal{C})$, its restriction to $\mathrm{Ch}_{[b,a]}$ is depicted as:

$$K_b \to K_{b-1} \to \cdots \to K_a.$$

Restrictions from below admits the following right adjoint:

$$(i_{\geq a})_*\colon\ \mathrm{Ch}_{\geq a}(\mathbf{Sp}) \to \mathrm{Ch}(\mathbf{Sp})$$

which given by padding with zeros:

$$(i_{\geq a})_* K(p) = \begin{cases} K(p) & p \geq a \\ 0 & \text{otherwise} \end{cases} \tag{14}$$

On the other hand, restriction from above have padding as left adjoint:

$$(i_{\leq b})_! K(p) = \begin{cases} 0 & p > b \\ K(p) & b \geq p \end{cases} \tag{15}$$

We denote the composition by:

$$K \mapsto K_{[b,a]} =: (i_{\leq b})_!(i_{\geq a})_* K$$

We can do the same for filtered object.

Definition 2.20. For two integers $a \leq b$ we define

$$\mathrm{Fil}_{[a,b]}(\mathbf{Sp}) := \mathrm{Fun}([a,b], \mathbf{Sp})$$

Note that the inclusion

$$i_{\leq n} : (-\infty, n] \to \mathbb{Z}$$

induces the restriction functor:

$$i^*_{\leq n} \colon \mathrm{Fil}(\mathbf{Sp}) \to \mathrm{Fil}_{(-\infty,n]}(\mathbf{Sp}) =: \mathrm{Fil}_{\leq n}(\mathbf{Sp})$$

and left adjoint

$$(i_{\leq n})_! \colon \mathrm{Fil}_{\leq n}(\mathbf{Sp}) \to \mathrm{Fil}(\mathbf{Sp})$$

which is given by

$$(i_{\leq n})_! F(p) = \begin{cases} F(p) & p \leq n \\ F(n) & \text{otherwise} \end{cases}$$

Similarly we define

$$\mathrm{Fil}_{\geq n}(\mathbf{Sp})$$

The map

$$i_{\geq n} : [n, \infty) \to \mathbb{Z}$$

induces the restriction

$$i^*_{\geq n} : \mathrm{Fil}(\mathbf{Sp}) \to \mathrm{Fil}_{\geq n}(\mathbf{Sp})$$

and it's right adjoint

$$(i_{\geq n})_* F(p) = \begin{cases} F(p) & n \leq p \\ 0 & \text{otherwise} \end{cases}$$

Note. sometimes we will identify between $F \in \widehat{\mathrm{Fil}}_{[n,m]}(\mathbf{Sp})$ to it's image in $\widehat{\mathrm{Fil}}(\mathbf{Sp})$. On the other hand we can define the "quotient truncation" functor.

Definition 2.21. Let $n \in \mathbb{Z}$ we define

$$q_n \colon \mathrm{Fil}(\mathbf{Sp}) \to \mathrm{Fil}_{\geq n}(\mathbf{Sp})$$

by

$$(q_n F)_p := \begin{cases} F_p / F_n & n+1 \leq p \\ 0 & \text{otherwise} \end{cases}$$

This functor behave nicely with the equivalence to chain complexes:

Corollary 2.22. *Let $F \in \widehat{\mathrm{Fil}}(\mathbf{Sp})$, we have that:*

$$\mathcal{A} q_n F \simeq i^*_{\geq n+1} \mathcal{A} F$$

treated as objects in $\mathrm{Ch}_{[\infty, n+1]}(\mathbf{Sp})$ or $\mathrm{Ch}(\mathbf{Sp})$.

Proof. A direct calculation shows that both are given by $(-)_p = \Sigma^{-p} \mathrm{gr}_p F$. □

Lemma 2.23. *Let $K \in \mathrm{Ch}(\mathbf{Sp})$ and $F = \mathcal{I}K$ its equivalent filtered spectrum, then*

$$\mathrm{Map}(\mathbb{S}_p^{-p}, i^*_{\geq n+1}K) \simeq F_p/F_n$$

Proof. Note that $\mathcal{A}F = K$. We use the corollary above and Eq. (3) to get:

$$\mathrm{Map}(\mathbb{S}_p^{-p}, i^*_{\geq n+1}K) \simeq \mathrm{Map}(\mathbb{S}_p^{-p}, \mathcal{A}q_nF) \simeq (q_nF)_p = F_p/F_n.$$

□

Let K be a chain complex, and let $F = \mathcal{I}K$, then by passing Lurie's construction of spectral sequence of F we get the following

Proposition 2.24. *Let $K \in \mathrm{Ch}(\mathbf{Sp})$ be a chain complex in Spectra, let $F = \mathcal{I}K$ the equivalent filtered spectrum and $E_r^{n,p}$ it's spectral sequence (see Definition 2.14 and Theorem 2.15). By passing Lurie's construction we get that*

$$E_1^{n,p} = \pi_{n-p}K_p,$$

Moreover, let $x\colon \mathbb{S}^{n-p} \to K_p$ represent an element in $E_1^{n,p}$. Then x survives to E_r if and only if there are maps of chain complexes:

$$\begin{tikzcd}
\mathbb{S}^{n-p} \arrow[r] \arrow[d, "x"'] & 0 \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & 0 \arrow[d] \\
K_p \arrow[r, "d_p"] & K_{p-1} \arrow[r, "d_{p-1}"] & \dots \arrow[r, "d_{p-r+2}"] & K_{p-r+1},
\end{tikzcd}$$

where the top row is concentrated in degree p, and the bottom row is the truncated chain complex from degree p to $p-r+1$.

Proof. The first page is given in Remark 2.18. Similar to $i_{\geq n}$, the inclusion

$$j \coloneqq i_{\geq p-1,p-r}\colon (\infty, p-1] \to (\infty, p-r]$$

induces the restriction functor j^*. By post composing we get a map

$$\mathrm{Map}(\mathbb{S}_p^{-p}, i^*_{\geq p-r+1}K) \xrightarrow{j^*\circ} \mathrm{Map}(\mathbb{S}_p^{-p}, i^*_{\geq p}K).$$

By suspending both sides (from $\mathbb{S}_p^{-p}$ to $\mathbb{S}_p^{n-p}$) we get the map

$$\mathrm{Map}(\mathbb{S}_p^{n-p}, i^*_{\geq p-r+1}K) \xrightarrow{j^*\circ} \mathrm{Map}(\mathbb{S}_p^{n-p}, i^*_{\geq p}K). \tag{16}$$

To get that visualization in the proposition, we first clarify that we use $\mathbb{S}_p^{n-p}$ for both the restricted and the usual versions of the chain that concentrated at p and we can write:

$$\begin{aligned}
&\mathrm{Map}_{[p,p-r+1]}(\mathbb{S}_p^{n-p}, K_{[p,p-r+1]}) \\
=&\mathrm{Map}_{[p,p-r+1]}(\mathbb{S}_p^{n-p}, i^*_{\leq p} i^*_{\geq p-r+1}K) \simeq \mathrm{Map}_{(\infty,p-r+1]}((i_{\leq p})_!\mathbb{S}_p^{n-p}, i^*_{\geq p-r+1}K) \\
=&\mathrm{Map}_{(\infty,p-r+1]}(\mathbb{S}_p^{n-p}, i^*_{\geq p-r+1}K) \simeq \Sigma^{n-p}F_p/F_{p-r}
\end{aligned}$$

where the qualities are notations and the last equivalence is Lemma 2.23. We can do the same to the right hand side of Eq. (16) and get the map:

$$\Sigma^{-n}F_p/F_{p-r} \simeq \mathrm{Map}(\mathbb{S}_p^{n-p}, K_{[p,p-r+1]}) \xrightarrow{j^*\circ} \mathrm{Map}(\mathbb{S}_p^{n-p}, K_p) \simeq \Sigma^{-n}F_p/F_{p-1} \tag{17}$$

By taking π_0 we conclude that $x \in \pi_n F_p/F_{p-1}$ is in the image of $\pi_n F_p/F_{p-r}$ if and only if the representing map

$$x : \mathbb{S}^n \to K_p$$

factor through $K_{[p,p-r+1]}$, which conclude the proof. □

Definition 2.25. A spectral sequence $\{E_r^{n,p}, d_r^{n,p}\}$ associated with a filtered object F converges conditionally to the homotopy groups of $\mathrm{colim}_p F_p$ if F is exhaustive and complete. Specifically, following Boardman [6, Def. 5.10], completeness requires that both the limit and the derived limit over the filtering vanish. In the ∞-categorical setting, this is equivalent to the single condition $\lim_p F_p \simeq 0$.

Remark 2.26. For any coherent chain complex K, the associated filtered object $F = \mathcal{I}K$ is complete by construction.

Corollary 2.27. *Let $K \in \mathrm{Ch}(\mathbf{Sp})$ be a coherent chain complex and $F = \mathcal{I}K$ its equivalent filtered spectrum.*

1. *The associated spectral sequence converges conditionally to $\pi_* \mathrm{colim}_p F_p$.*

2. *If K is bounded below, then the spectral sequence converges strongly to $\pi_* \mathrm{colim}_p F_p$.*

Proof. For the first statement, since F is complete ($\lim_p F_p \simeq 0$), the Milnor exact sequence applied to the sequential limit of F_p implies that the $\lim^1$ term also vanishes. This satisfies the precise criteria for conditional convergence [6, Def. 5.10]. For the second statement, if K is bounded below, then F is a bounded below filtered object. By Theorem 2.15, this bounded below condition promotes conditional convergence to strong convergence. □

Example 2.28. Returning to Example 2.16, consider an element $x\colon \mathbb{S}^{n-p} \to K_p$ representing a class in $E_1^{n,p}$. The differential $d_1^{n,p}\colon E_1^{n,p} \to E_1^{n-1,p-1}$ is given by:

$$d_1^{n,p}(x) = d_p \circ x, \tag{18}$$

where $d_p\colon K_p \to K_{p-1}$ is the differential in the chain complex K.

By Proposition 2.24, x survives to the second page if and only if $d_p \circ x$ is null-homotopic, i.e., x is a cycle:

$$d_p \circ x = 0 \quad \text{in} \quad \pi_{n-p}K_{p-1}. \tag{19}$$

Similarly, x survives to the r-th page if it extends to a map of chain complexes over r steps, corresponding to being a cycle up to r-th order.

Let F be a filtered object in $\mathcal{C}$. Recall from Definition 2.14 that we have the spectral sequence $\{E_r^{n,p}, d_r^{n,p}\}$ associated with F. To understand the differentials $d_r^{n,p}$, we consider the following commutative diagram arising from the connecting morphisms in the long exact sequences of homotopy groups:

$$\begin{tikzcd}
\pi_n\left({}^{F_p}/{}_{F_{p-r}}\right) \arrow[r] \arrow[d, "\delta_{p-r}"'] & \pi_n\left({}^{F_p}/{}_{F_{p-1}}\right) \arrow[r] & \pi_n\left({}^{F_{p+r-1}}/{}_{F_{p-1}}\right) \arrow[d, "\delta_{p-1}"] \\
\pi_{n-1}\left({}^{F_{p-r}}/{}_{F_{p-2r}}\right) \arrow[r] & \pi_{n-1}\left({}^{F_{p-r}}/{}_{F_{p-r-1}}\right) \arrow[r] & \pi_{n-1}\left({}^{F_{p-1}}/{}_{F_{p-r-1}}\right),
\end{tikzcd} \tag{20}$$

where the vertical maps δ_{p-r} and δ_{p-1} are the connecting homomorphisms from the long exact sequences associated with the cofiber sequences in the filtration.

Let $x \in E_r^{n,p}$ be an element that survives to the r-th page, and let $x' \in \pi_n\left({}^{F_p}/{}_{F_{p-r}}\right)$ be a lift of x. By the commutativity of Diagram Equation (20), the image of $\delta_{p-r}(x')$ in $\pi_{n-1}\left({}^{F_{p-1}}/{}_{F_{p-r-1}}\right)$ depends only on x and not on the specific choice of the lift x'. This property allows us to define the differential $d_r^{n,p}$.

Definition 2.29. Let F be a filtered spectrum with spectral sequence $E_r^{n,p}$ and let $x \in E_r^{n,p}$. Let $x' \in \pi_n\left({}^{F_p}/{}_{F_{p-r}}\right)$ be a lift of x. We define the differential

$$d_r^{n,p}(x) \in E_r^{n-1,p-r}$$

to be the image of $\delta_{p-r}(x')$ in $E_r^{n-1,p-r}$ under the natural map induced by the composition

$$\pi_{n-1}\left({}^{F_{p-r}}/{}_{F_{p-2r}}\right) \to E_r^{n-1,p-r}.$$

Diagrammatically, this is represented as:

$$\begin{tikzcd}
x' \arrow[r] \arrow[d, "\delta_{p-r}"'] & x \\
\delta_{p-r}(x') \arrow[r, mapsto] & d_r^{n,p}(x).
\end{tikzcd} \tag{21}$$

The definition ensures that $d_r^{n,p}(x)$ is well-defined, independent of the choice of lift x'.

Chain-level description of the differential

Similar to the description of the pages, as maps between chain complexes, we can unwind Lurie's description for filtered spectrum, to get a chain complex description of the differential.

To facilitate the proof of Proposition 2.32, we first introduce the following definition and lemma.

Definition 2.30. We denote the "shift to the left" functor by

$$T\colon \operatorname{Ch}(\mathbf{Sp}) \to \operatorname{Ch}(\mathbf{Sp})$$

which is given by precomposing with

$$\text{-}1 : \mathbb{Z} \to \mathbb{Z}, \quad n \mapsto n-1$$

Lemma 2.31. *Let $L \in \operatorname{Ch}_{(\infty,n]}(\mathbf{Sp})$ and $K \in \operatorname{Ch}_{[n,-\infty)}(\mathbf{Sp})$ be two chain complexes. Then*

$$\operatorname{Map}(\Sigma^{-1}TL, K) \simeq \operatorname{Map}(L, K)$$

Proof. First, note that $\widehat{\mathrm{Fil}}(\mathbf{Sp})$ is generated (under colimits) by よ_p. Passing to $\mathrm{Ch}(\mathbf{Sp})$, we get that $\mathrm{Ch}(\mathbf{Sp})$ is generated by $\mathcal{I}\,\text{よ}_p = \mathbb{S}_p^{-p}$. Using this observation, it suffices to show that:

$$\mathrm{Map}(\mathbb{S}_p^{-p}, K) \simeq \mathrm{Map}(\Sigma^{-1}T\mathbb{S}_p^{-p}, K) \simeq \mathrm{Map}(\mathbb{S}_{p+1}^{-(p+1)}, K)$$

and by taking $\mathcal{I}$, our goal is to show:

$$\mathrm{Map}(\text{よ}_p, \mathcal{I}K) \simeq \mathrm{Map}(\text{よ}_{p+1}, \mathcal{I}K)$$

which is equivalent to

$$(\mathcal{I}K)_p \simeq (\mathcal{I}K)_{p+1} \tag{22}$$

Note that the assumption on L translates to $p \geq n$.

Let $F = \mathcal{I}K$, where $K \in \mathrm{Ch}_{[n,-\infty)}(\mathbf{Sp})$. Then for $q \geq n+1$ we have

$$0 = K_q = (\mathcal{A}F)_q = \Sigma^{-q}F_q/F_{q-1}$$

which shows that Eq. (22) holds. □

Example. To get the direction right, note that $T\mathbb{S}_0 = \mathbb{S}_1$ and get that:

$$\mathrm{Map}(\mathbb{S}_1, \mathbb{S}_0) \simeq \mathrm{Map}(\Sigma^{-1}T\mathbb{S}_0^1, \mathbb{S}_0) \simeq \mathrm{Map}(\mathbb{S}_0^1, \mathbb{S}_0).$$

So shifting the sourse to the right shifts the homotopical degree of the source one up.

Proposition 2.32. *Let $K \in \mathrm{Ch}(\mathbf{Sp})$ be a chain complex in Spectra, let $F = \mathcal{I}K$ be its associated filtered spectrum, and let $E_r^{n,p}$ be the spectral sequence of F. Let $x \in \mathcal{E}_r^{n,p}$ be an element surviving to the r-th page (see Definition 2.17). Then the differential $\partial_r(x)$ is given by the composition of chain maps:*

$$\begin{tikzcd}
\mathbb{S}^{n-p} \arrow[r] \arrow[d, "x"'] & \dots \arrow[r] \arrow[d] & 0 \arrow[d] \\
K_p \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & K_{p-r+1} \arrow[d] \\
0 \arrow[r] & \dots \arrow[r] & K_{p-r}
\end{tikzcd}$$

viewed as a map $\mathbb{S}^{n-p} \to K_{p-r}$, where the first map is an extension of x (see Proposition 2.24), and the second is induced by the structure of the chain complex K.

Proof. We begin with Lurie's description, Eq. (10). The differential is given by the map:

$$\pi_n F_p/F_{p-r} \to \pi_{n-1}F_{p-r}/F_{p-2r} \tag{23}$$

hence ∂_r is induced by post-composing with the connecting morphism:

$$F_p/F_{p-r} \to \Sigma F_{p-r}/F_{p-2r}$$

which comes from the cofiber sequence:

$$F_{p-r}/F_{p-2r} \to F_p/F_{p-2r} \to F_p/F_{p-r}.$$

As we saw in the proof of Proposition 2.24, we can use Lemma 2.23 to see that the map $F_p/F_{p-2r} \to F_p/F_{p-r}$ is representable as

$$j^*\colon \operatorname{Map}(\mathbb{S}_p^{-p}, K_{[p,p-2r+1]}) \xrightarrow{j\circ} \operatorname{Map}(\mathbb{S}_p^{-p}, K_{[p,p-r+1]})$$

which is given by postcomposing with restriction and padding. By taking the cofiber of j, we get the cofiber of j^*. The cofiber is calculated pointwise: for $q \in [p-r, p-2r+1]$, the map j^* is $K_q \to 0$, and for $q \notin [p-r, p-2r+1]$, the map is the identity. We conclude that the cofiber map is of the form:

$$c\colon K_{[p,p-r+1]} \to \operatorname{cof}(j^*) \simeq \Sigma K_{[p-r,p-2r+1]}.$$

Hence, the differential is represented by post-composing:

$$c^*\colon \operatorname{Map}(\mathbb{S}_p^{-p}, K_{[p,p-r+1]}) \xrightarrow{c\,\circ} \operatorname{Map}(\mathbb{S}_p^{-p}, \Sigma K_{[p-r,p-2r+1]}).$$

Note that $\mathbb{S}_p^{-p} \in \mathrm{Ch}_{(\infty,p]}(\mathbf{Sp})$ and the other two chains are elements in $\mathrm{Ch}_{[p,-\infty)}(\mathbf{Sp})$. So we can use Lemma 2.31 on c and find that c^* is equivalent to post-composition with

$$K_{[p,p-r+1]} \to T^{-1}K_{[p-r,p-2r+1]}$$

which is the visualization in the proposition. Note that the diagram (which has K_{p-r} as a target) is a visualization of $\partial_r(x) \in E_1^{n-1,p-r}$, and the "non-truncated" version (with target $K_{[p-r,p-2r+1]}$) is the evidence that we can lift $\partial_r(x)$ to $\mathcal{E}_r^{n-1,p-r}$.

Let $K = \mathcal{A}F$, we get a cofiber sequence of chain maps:

$$\begin{tikzcd}
0 \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & K_{p-r} \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & K_{p-2r+1} \arrow[d] \\
K_p \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & K_{p-r+1} \arrow[r] \arrow[d] & K_{p-r} \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & K_{p-2r+1} \arrow[d] \\
K_p \arrow[r] & \dots \arrow[r] & K_{p-r+1} \arrow[r] & 0 \arrow[r] & \dots \arrow[r] & 0
\end{tikzcd}$$

Colimits in the functor category of chains are computed pointwise, hence the connecting morphism is given by

$$\begin{tikzcd}
K_p \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & K_{p-r+1} \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & 0 \arrow[d] \\
0 \arrow[r] & \dots \arrow[r] & 0 \arrow[r] & \Sigma K_{p-r} \arrow[r] & \dots \arrow[r] & \Sigma K_{p-2r+1}
\end{tikzcd} \tag{24}$$

We denote the restriction of chains:

$$K_{[p,p-r+1]} := K_p \to \cdots \to K_{p-r+1} \to \overbrace{0 \to \cdots \to 0}^{r-1}$$
$$K_{[p-r,p-2r+1]} := \overbrace{0 \to \cdots \to 0}^{r-1} \to K_{p-r} \to \cdots \to K_{p-2r+1}$$

as objects in $\mathrm{Ch}_{[p,p-2r+2]}(\mathbf{Sp})$. The diagram Eq. (24) is equivalent to the diagram:

$$\begin{tikzcd} K_{[p,p-r+1]} \arrow[r] \arrow[d] & 0 \arrow[d] \\ 0 \arrow[r] & \Sigma K_{[p-r,p-2r+1]} \end{tikzcd}$$

Using the stability of $\mathbf{Sp}$, we get that $\mathrm{Ch}_{[p,p-2r+2]}(\mathbf{Sp})$ is stable, and the last diagram is equivalent to the map of chains: $K_{[p,p-r+1]} \to K_{[p-r,p-2r+1]}$ which can be drawn as:

$$\begin{tikzcd} K_p \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & K_{p-r+1} \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & 0 \arrow[d] \\ 0 \arrow[r] & \dots \arrow[r] & K_{p-r} \arrow[r] & \dots \arrow[r] & K_{p-2r+1} \end{tikzcd}$$

Hence, pre-composition with a lift of x to a map $x' : \mathbb{S}_p^{n-p} \to K_{[p,p-r+1]}$ gives the diagram:

$$\begin{tikzcd} \mathbb{S}^{n-p} \arrow[r] \arrow[d, "x"'] & \dots \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & 0 \arrow[d] \\ K_p \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & K_{p-r+1} \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & 0 \arrow[d] \\ 0 \arrow[r] & \dots \arrow[r] & K_{p-r} \arrow[r] & \dots \arrow[r] & K_{p-2r+1} \end{tikzcd} \tag{25}$$

and the truncated version:

$$\begin{tikzcd} \mathbb{S}^{n-p} \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & 0 \arrow[d] \\ K_p \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & K_{p-r+1} \arrow[d] \\ 0 \arrow[r] & \dots \arrow[r] & K_{p-r} \end{tikzcd}$$

The last diagram is $\partial_r(x) \in E_1^{n-1,p-r}$, and the previous diagram Eq. (25) is the evidence that we can lift $\partial_r(x)$ to $\mathcal{E}_r^{n-1,p-r}$. □

Theorem B. *Let $K \in \mathrm{Ch}(\mathbf{Sp})$ be a coherent chain complex in Spectra. Then there is a spectral sequence:*

$$E_1^{n,p} = \pi_{n-p} K_p,$$

which conditionally converges to

$$E_r^{n,p} \implies \pi_n \operatorname*{colim}_p (\mathcal{I}K)_p,$$

Moreover, let $x : \mathbb{S}^{n-p} \to K_p$ represent an element of $E_1^{n,p}$. Then x survives to $E_{r+1}^{n,p}$ if and only if there exists a chain complex map:

$$\begin{tikzcd} \mathbb{S}^{n-p} \arrow[r] \arrow[d, "x"'] & 0 \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & 0 \arrow[d] \\ K_p \arrow[r, "d_p"] & K_{p-1} \arrow[r, "d_{p-1}"] & \dots \arrow[r, "d_{p-r}"] & K_{p-r} \end{tikzcd} \tag{2}$$

Here, the top row is concentrated in degree p, and the bottom row is the restriction of K from degree p down to $p-r$. The differential $\partial_{r+1}(x)$ is given by the composition of chain maps:

$$\begin{tikzcd}
\mathbb{S}^{n-p} \arrow[r] \arrow[d, "x"'] & \dots \arrow[r] \arrow[d] & 0 \arrow[d] \\
K_p \arrow[r] \arrow[d] & \dots \arrow[r] \arrow[d] & K_{p-r} \arrow[d] \\
0 \arrow[r] & \dots \arrow[r] & K_{p-r-1}
\end{tikzcd}$$

The first map is precisely Eq. (2) and the second is induced by the structure of the chain complex K.

3 Manifolds with Corners and Framed Flow Categories

This section lays the geometric and categorical groundwork for our main constructions. We begin by introducing a diagrammatic framework. The fundamental geometric objects in this setting are manifolds with corners, which generalize smooth manifolds (see [12] for a relevant treatment). These concepts are then used to define flow categories [8], where the morphism spaces are manifolds with corners and composition is given by embedding into the corner strata. The subsequent subsections develop the precise definitions required for this approach.

3.1 Diagrammatic framework

Let A be a set and $\mathcal{C}$ a category. An $\langle A\rangle$ diagram in $\mathcal{C}$ is a functor:

$$F\colon \{1 \to 0\}^A \to \mathcal{C}.$$

In the case where $C \in \mathcal{C}$ have a name (e.g. space, manifold, group), we call such functor an $\langle A\rangle$ space, etc.

Definition 3.1. Let A be a finite set and $a \in A$ then we define its **A-boundary** to be the restriction along the inclusion

$$i : \mathrm{Fun}(A \setminus \{a\}, \{1 \to 0\}) \to \mathrm{Fun}(A, \{1 \to 0\})$$

where i is given by taking the full subcategory spanned by functors with $F(a) = 1$. We get a a map:

$$\partial_a\colon \mathrm{Fun}(\{1 \to 0\}^A, \mathcal{C}) \to \mathrm{Fun}(\{1 \to 0\}^{A\setminus\{a\}}, \mathcal{C})$$

An $\langle A\rangle$**-map** is a natural transformation between $\langle A\rangle$-diagrams.

Remark 3.2. Notice that

$$\mathrm{Fun}(1, \mathrm{Fun}(\{1 \to 0\}^A, \mathcal{C})) \simeq \mathrm{Fun}(\{1 \to 0\}^A, \mathrm{Fun}(1, \mathcal{C}))$$

hence any $\langle A\rangle$-map is an $\langle A\rangle$ diagram in the arrow category $\mathrm{Fun}(1, \mathcal{C})$, and we can use $\langle A\rangle$ diagrams notations, like

$$\partial_a \mu$$

Definition 3.3. Let $\mathcal{C}$ be a monoidal category. For $i \in \{1,2\}$, let A_i be finite sets and F_i an $\langle A_i \rangle$ diagram. Then there is a tensor product

$$F_1 \otimes F_2(B_1 \sqcup B_2) \coloneqq F_1(B_1) \otimes F_2(B_2).$$

which is an $\langle A_1 \sqcup A_2 \rangle$ diagram in $\mathcal{C}$. Given two morphisms $\mu_i \colon F_i \to G_i$ we define

$$(\mu_1 \otimes \mu_2)(B_1 \sqcup B_2) \coloneqq \mu_1(B_1) \otimes \mu_2(B_2)$$

which is a map of $\langle A_1 \sqcup A_2 \rangle$ diagrams.

Taking $\mathcal{C} = \mathbf{Top}$ and let A be a finite set, here are few useful constructions that we will use later. We remind the reader that the one point compactification is a monoidal functor from locally compact Hausdorff spaces with proper maps with disjoint union (resp. Cartesian product) to pointed spaces with wedge sum (resp. smash product). Moreover it is a contravariant functor from the category of topological spaces with open inclusions, also known as the Pontrjagin-Thom collapse map. Explicitly, given $f : X \to Y$ and map continuous map between topological spaces the collapse map is the map $Y_+ \to X_+$

$$y \mapsto \begin{cases} x & y = f(x) \\ \infty_X & \text{otherwise} \end{cases}$$

Let X be $\langle A \rangle$ diagram in $\mathbf{Top}$ such that all the maps are proper. We have an $\langle A \rangle$ diagram, X_+, given by

$$X_+(B) \coloneqq (X(B))_+$$

which arising from the functoriality.

Definition 3.4. Let $f : X \to Y$ be a map of $\langle A \rangle$ diagrams in $\mathbf{Top}$ such that $f(B)$ is open inclusion, then the functoriality of $(-)_+$ yields the **collapse map**

$$f_+ : Y_+ \to X_+$$

Definition 3.5. Let A be a finite set and M, N two $\langle A \rangle$ manifolds, we say that a map $e : M \to N$ is **neat embedding** if

1. For each $B \subseteq A$, the map $e(B) \colon M(B) \to N(B)$ is an embedding.
2. The intersections $M(B) \cap N(C)$ are transverse for all $B \subseteq C$.
3. The preimage of the faces satisfies $e^{-1}(\partial_a N) = \partial_a M$ for all $a \in A$.

For $i \in \{1,2\}$ let A_i be finite sets and let $e_i : M_i \to N_i$ be an two neat embeddings of $\langle A \rangle_i$ diagrams of manifolds. Taking the monoidal structure $(\mathbf{Top}, \times)$ we can use Definition 3.3 and get a map

$$e_1 \times e_2 \colon M_1 \times M_2 \to N_1 \times N_2$$

which is a map of $\langle A_1 \sqcup A_2 \rangle$ diagrams.

Moreover, we can define the **normal bundle**, ν_{e_i}, by

$$(\nu_{e_i})(B) \coloneqq \nu_{e_i}(B)$$

which are $\langle A \rangle$ diagrams in **Top** of the total spaces of the bundles. We can do the same for the map $e_1 \times e_2$ and the fact that the total space of normal bundle commute with Cartesian product of the base, yields:

$$\nu_{e_1} \times \nu_{e_2} = \nu_{e_1 \times e_2}$$

as $\langle A_1 \sqcup A_2 \rangle$ diagrams.

Lastly, we have open inclusions of diagrams:

$$\nu_{e_i} \to N_i, \quad \nu_{e_1} \times \nu_{e_2} \to N_1 \times N_2$$

hence we can compactify and get maps

$$(N_i)_+ \to (\nu e_i)_+, \quad (N_1 \times N_2)_+ \to (\nu_{e_1} \times \nu_{e_2})_+$$

and the monoidality of $(-)_+$ give rise to a map of $\langle A_1 \sqcup A_2 \rangle$ diagrams:

$$(N_1)_+ \wedge (N_1)_+ \to (\nu_{e_1})_+ \wedge (\nu_{e_2})_+.$$

Definition 3.6. Given an $\langle A \rangle$ vector bundle over an $\langle A \rangle$ manifold, i.e. an $\langle A \rangle$ map $\pi : E \to M$, such that $\pi(B)$ is vector bundle for $A \subseteq A$, a **trivialization** of E is a map of $\langle A \rangle$-spaces

$$\varphi : E \simeq M \times \mathbb{R}^N$$

for some $N \in \mathbb{N}$, where $\mathbb{R}^N$ is the $\langle \varnothing \rangle$ diagram on $\mathbb{R}^N$. This map amounts to family of bundle maps

$$\varphi(B) : E(B) \simeq M(B) \times \mathbb{R}^N.$$

We can project to the second coordinate of the RHS and get a map

$$E \to \underline{\mathbb{R}^N},$$

where the RHS is the constant functor on $\mathbb{R}^N$. Applying $(-)_+$ resulting an $\langle A \rangle$-map

$$E_+ \to \underline{(\mathbb{R}^N)_+} \simeq \underline{S^N}. \tag{26}$$

Our main application is the case where we have a neat embedding of $\langle A \rangle$ manifolds

$$e : M \hookrightarrow \mathbb{R}^N \times \mathbb{R}^A_{0\leq}$$

and a framing of the normal bundle

$$\varphi : \nu_e \simeq M \times \mathbb{R}^{N+|A|-n}.$$

We construct the **Thom map**:

$$f_{e,\varphi} : (\mathbb{R}^N \times \mathbb{R}^A_{0\leq})_+ \to (\nu_e)_+ \to (\mathbb{R}^{N+|A|-n})_+$$

where the first map is the collapse map of e and the second map is the projection of φ to the second coordinate, and applying $(-)_+$.

If $|A| = n$ we ca write

$$f : S^N \wedge (\mathbb{R}^A_{0\leq})_+ \to (\nu_e)_+ \to S^N. \tag{27}$$

3.2 Manifolds with corners

Let $\mathbb{R}_{0\leq}$ denote the half-line:

$$\mathbb{R}_{0\leq} := [0, \infty).$$

Definition 3.7. An n-dimensional **manifold with corners** is a topological space M toghter with an atlas, modeled on open subsets of $\mathbb{R}^n_{0\leq}$ with smooth transition functions. We define $c(m)$ as the number of coordinates equal to zero in a chart around m. A **connected face** of M is a closures of connected components of:

$$\{m \in M \mid c(m) = 1\}.$$

A **face** is a union of connected faces.

An n-dimensional **manifolds with faces** is an n-dimensional manifold with corners such that every $m \in M$ belong to $c(m)$ connected faces of M.

Note that a face of an n-dimensional manifolds with faces is an $(n-1)$-dimensional manifolds with faces.

Definition 3.8. Let A be a finite set. An $\langle A\rangle$**-manifold** is a manifold with faces M, together with a collection of faces $\{\partial_a M\}_{a\in A}$ satisfying the following conditions:

1. The union of all faces covers the boundary of M:

$$\bigcup_{a\in A} \partial_a M = \partial M.$$

2. For $a \neq b$, the intersection $\partial_a M \cap \partial_b M$ is a face of both $\partial_a M$ and $\partial_b M$.

If $A = \{1, \dots, k\}$, we refer to M as a $\langle k\rangle$**-manifold**.

Note. Using the definitions of $\langle A\rangle$-manifold, we get that $\partial_a M$ inherit a structure of an $\langle A \setminus a\rangle$-manifold.

Example 3.9. The space $\mathbb{R}^A_{0\leq}$ admits an $\langle A\rangle$-manifold structure by defining the faces:

$$\partial_a \mathbb{R}^A_{0\leq} = \mathbb{R}^{A\setminus\{a\}}_{0\leq} = \{x \in \mathbb{R}^A_{0\leq} \mid x_a = 0\}.$$

More generally, $\mathbb{R}^N \times \mathbb{R}^A_{0\leq}$ can be given an $(N + |A|)$-dimensional $\langle A\rangle$-manifold structure by setting:

$$\partial_a(\mathbb{R}^N \times \mathbb{R}^A_{0\leq}) = \mathbb{R}^N \times \partial_a \mathbb{R}^A_{0\leq}.$$

Example 3.10. Consider a triangle in the plane. It is a 2-dimensional manifold with corners, which admits a $\langle 3\rangle$-manifold structure by taking the three edges as the faces $\partial_a M$. This example illustrates that not all k-manifolds are necessarily k-dimensional. However, in this paper, we focus on manifolds where the dimension equals the number of faces, i.e., $n = k$.

Construction 3.11. Let M be an n-dimensional $\langle A\rangle$-manifold. It induces an $\langle A\rangle$-diagram in the category of manifolds with embeddings by defining:

$$M(B) = \bigcap_{b\in B} \partial_b M,$$

for each subset $B \subseteq A$. In particular, $M(\varnothing) = M$, and $M(B)$ is an $(n - |B|)$-dimensional $\langle B\rangle$-manifold.

Remark. For simplicity, we sometimes refer to an $\langle A\rangle$-manifold by its associated $\langle A\rangle$-diagram.

Example 3.12. The manifold $\mathbb{R}^3_{0\leq}$ induces the following $\langle 3\rangle$-diagram:

$$\begin{tikzcd}
& \mathrm{pt} \arrow[dl] \arrow[d] \arrow[dr] & \\
\mathbb{R}^{\{z\}}_{0\leq} \arrow[d] \arrow[dr] & \mathbb{R}^{\{y\}}_{0\leq} \arrow[dl] \arrow[dr] & \mathbb{R}^{\{z\}}_{0\leq} \arrow[d] \arrow[dl] \\
\mathbb{R}^{\{y,z\}}_{0\leq} \arrow[dr] & \mathbb{R}^{\{x,z\}}_{0\leq} \arrow[d] & \mathbb{R}^{\{x,y\}}_{0\leq} \arrow[dl] \\
& \mathbb{R}^{\{x,y,z\}}_{0\leq} &
\end{tikzcd}$$

The top vertex represents the origin. The second row consists of the coordinate axes $\mathbb{R}^{\{x\}}_{0\leq}$, $\mathbb{R}^{\{y\}}_{0\leq}$, and $\mathbb{R}^{\{z\}}_{0\leq}$. The third row contains the coordinate planes, which correspond to the faces $\partial_a\mathbb{R}^3_{0\leq}$. The bottom vertex is the entire manifold $\mathbb{R}^3_{0\leq}$.

Construction 3.13. For $i \in \{1,2\}$, let M_i be an n-dimensional $\langle A\rangle_i$-manifold, then the Cartesian product $M_1 \times M_2$ admits an $\langle A_1 \sqcup A_2\rangle$-manifold structure by

$$\partial_a(M_1 \times M_2) \coloneqq \begin{cases} (\partial_a M_1) \times M_2 & a \in A_1 \\ M_1 \times (\partial_a M_2) & a \in A_2 \end{cases}$$

Note that this construction coincides with Definition 3.3.

Definition 3.14. A neat embedding of $\langle A\rangle$-manifolds with corners is a map of topological spaces $e : M \to N$ such that the induces map of $\langle A\rangle$-diagrams is neat embedding.

Definition 3.15. Let M be an n dimensional $\langle A\rangle$ manifold, $e : M \hookrightarrow \mathbb{R}^N \times \mathbb{R}^A_{0\leq}$, a neat embedding. Note that the normal bundle is an $\langle A\rangle$ vector bundle over M, of rank $N + |A| - n$. A **normal framing** is a trivialization (see Definition 3.6) of ν_e

$$\varphi : \nu_e \simeq M \times \mathbb{R}^{N+|A|-n}.$$

We denote by

$$Th(M, e) \coloneqq (\nu_e)_+$$

the **Thom space** of M and e, which is an $\langle A\rangle$ space.

If the choice of neat embedding is clear, we will omit e and write $Th(M)$.

Remark. Let n be the dimension of M, each normal bundle $\nu_e M(B)$ is of rank $N + |A| - n$.

Similar to Construction 3.13, for $i \in \{1,2\}$, let M_i be an embedded manifold with normal framing φ_i. Since the normal bundle of a Cartesian product is the direct sum of the individual normal bundles, the Cartesian product $M_1 \times M_2$ admits a normal framing. We denote the resulting framing by $\varphi_1 \oplus \varphi_2$.

3.3 Framed Flow Categories

We now introduce framed flow categories, first developed by Cohen, Jones, and Segal [8] to axiomatize the geometric data arising from Morse-Smale functions and certain Floer-theoretic setups. A flow category is a graded category whose morphism spaces are manifolds with corners, with dimensions determined by the grading. Composition is realized geometrically by identifying the boundary strata of one morphism manifold with the products of others.

A *framed* flow category further equips these morphism manifolds with a stable framing that must be compatible with the composition law. This coherent structure is what allows for the construction of a stable homotopy type. Our treatment, which relies on neat embeddings of these manifolds into Euclidean space, follows the perspective used in modern applications like [14].

Definition 3.16. Let $\mathscr{C}$ be a category with a **grading function**:

$$|-|\colon \operatorname{ob}(\mathscr{C}) \to \mathbb{Z}.$$

For each pair of objects $x, y \in \mathscr{C}$, define the linearly ordered set:

$$J_{y,x} \coloneqq \{i \in \mathbb{Z} \mid |y| > i > |x|\}. \tag{28}$$

Definition 3.17. A **flow category** is a topologically enriched category $\mathscr{C}$ with a discrete set of objects and a grading function

$$|-| : \operatorname{ob}(\mathscr{C}) \to \mathbb{Z}$$

Moreover, for each pair $x, y \in \mathcal{C}$ the morphism space $\operatorname{Hom}_{\mathscr{C}}(y, x)$ is endowed with a structure of an $|J_{y,x}|$-dimensional $\langle J_{y,x}\rangle$-manifold, also denoted by $M_{y,x}$, such that the corner structure satisfies:

$$\partial_i M_{z,x} = \bigsqcup_{y\in \operatorname{ob}(\mathscr{C}),\ |y|=i} M_{z,y} \times M_{y,x}, \tag{29}$$

for all $i \in J_{z,x}$. The composition is given by the inclusion of the boundary:

$$M_{z,y} \times M_{y,x} \hookrightarrow \partial_i M_{z,x} \hookrightarrow M_{z,x}.$$

Notation 3.18. Let $\mathscr{C}$ be a flow category and $z, y, x \in \mathscr{C}$, we will write $\partial_y M_{z,x}$ to denote $M_{z,y} \times M_{y,x}$ as connected component of the the face $\partial_i M_{z,x}$.

Notation 3.19. Let $\mathscr{C}$ be a flow category and $j, i \in \mathbb{Z}$ we denote

$$M_{i,j} \coloneqq \bigsqcup_{\substack{y\in\mathscr{C},|y|=j \\ x\in\mathscr{C},|x|=i}} M_{y,x}$$

Example 3.20. Consider a framed flow category $\mathscr{C}$ with object set $\{z, y, x, w\}$ and grading $|z| = 4, \dots |w| = 1$. The morphism spaces are:

- Zero-dimensional manifolds $M_{z,y}$, $M_{y,x}$, $M_{x,w}$, which are discrete points (possibly with signs).
- One-dimensional framed manifolds $M_{z,x}$ and $M_{y,w}$, serving as null-bordisms:

$$\partial_3 M_{z,x} = M_{z,y} \times M_{y,x}, \quad \partial_2 M_{y,w} = M_{y,x} \times M_{x,w}.$$

- A two-dimensional manifold $M_{z,w}$ with faces:

$$\partial_3 M_{z,w} = M_{z,y} \times M_{y,w}, \quad \partial_2 M_{z,w} = M_{z,x} \times M_{x,w},$$

and corners given by $M_{z,y} \times M_{y,x} \times M_{x,w}$.

Definition 3.21. A *neat embedding* of a flow category $\mathscr{C}$ consists of a choice of integer N and a family of neat embeddings of manifolds with corners,

$$e_{y,x} \colon M_{y,x} \hookrightarrow \mathbb{R}^{N(|y|-|x|)} \times \mathbb{R}^{J_{y,x}}_{0\leq},$$

for each pair of objects y, x, such that the following diagram of $\langle J_{z,x} \setminus \{|y|\}\rangle$-spaces, commutes

$$\begin{tikzcd}
M_{z,y} \times M_{y,x} \arrow[r, "e_{z,y} \times e_{y,z}"] \arrow[d] & \mathbb{R}^{N(|z|-|y|)} \times \mathbb{R}^{J_{z,y}}_{0\leq} \times \mathbb{R}^{N(|y|-|x|)} \times \mathbb{R}^{J_{y,x}}_{0\leq} \arrow[d] \\
\partial_y M_{z,x} \arrow[r, "\partial_i e_{z,x}"] & \mathbb{R}^{N(|z|-|x|)} \times \partial_i \mathbb{R}^{J_{z,x}}_{0\leq}
\end{tikzcd}$$

where $i = |y|$, the bottom map is the restriction of $\partial_i e_{z,x}$ to the y component the top map is Definition 3.3 and the vertical maps are diffeomorphisms.

Note that is a uniform version, in the sense that all the manifolds embeds into a products of $\mathbb{R}^N$. This version arises by stabilizing [12].

Remark 3.22. The fact that any manifold with corners admits a neat embedding implies that any flow category can be embedded, see [12] Lemma 3.16. Moreover, the space of such embeddings becomes arbitrarily connected as N increases.

do we need this? (wait for index bundle change first)

We recall that for two bundles $E_i \to M_i$, we have the following.

$$Th(E_1 \boxtimes E_2) \simeq Th(E_1) \wedge Th(E_2)$$

where the LHS is the Thom space of the bundle over $M_1 \times M_2$. Using the functoriality of the collapse map (Definition 3.4,Definition 3.4), and the fact above, we get a commutative diagram of $\langle A\rangle$-spaces

$$\begin{tikzcd}
S^{N(|z|-|x|)} \wedge (\mathbb{R}^{J_{z,y}}_{0\leq} \times \mathbb{R}^{J_{y,x}}_{0\leq})_+ \arrow[r] \arrow[d] & Th(M_{z,y}) \wedge Th(M_{y,x}) \arrow[r, "\simeq"] & Th(M_{z,y} \times M_{y,x}) \arrow[d] \\
S^{N(|z|-|x|)} \wedge (\partial_i \mathbb{R}^{J_{z,x}}_{0\leq})_+ \arrow[rr] & & \partial_y Th(M_{z,x})
\end{tikzcd} \tag{30}$$

Definition 3.23. Let $\mathscr{C}$ be an embedded flow category. A normal **framing** of $\mathscr{C}$ is a choice of normal framing

$$\varphi_{y,x} \colon \nu M_{y,x} \simeq M_{y,x} \times \mathbb{R}^{N(|y|-|x|)}$$

for each two objects $x, y \in \mathcal{C}$, with the following coherency

$$\partial_y \varphi_{z,x} = \varphi_{z,y} \oplus \varphi_{y,x}.$$

The compatibility of the neat embedding and the framing implies the commutativity of the diagram:

$$\begin{tikzcd} S^{N(|z|-|x|)} \wedge (\mathbb{R}^{J_{z,y}}_{0\le} \times \mathbb{R}^{J_{y,x}}_{0\le})_+ \arrow[r] \arrow[d] & Th(M_{z,y} \times M_{y,x}) \arrow[r, "\varphi_{z,y}\oplus\varphi_{y,x}"] \arrow[d, "\simeq"'] & \underline{S^{N(|z|-|y|)}} \wedge \underline{S^{N(|y|-|x|)}} \arrow[d, "\mathrm{id}"] \\ S^{N(|z|-|x|)} \wedge (\partial_i \mathbb{R}^{J_{z,x}}_{0\le})_+ \arrow[r] & Th(\partial_y M_{z,x}) \arrow[r, "\partial_y \varphi_{z,x}"] & \underline{S^{N(|z|-|x|)}}. \end{tikzcd} \tag{31}$$

where the vertical maps are the "Thom" maps; see Eq. (27).

3.4 Flow Modules

Definition 3.24. Let $\mathscr{C}$ be a flow category. A **right flow module** W of **degree** n over $\mathscr{C}$ consists of a collection of manifolds with corners $\{W_x\}_{x\in\mathscr{C}}$ together with compatible corner structure maps (embeddings of codimension 1 boundary faces):

$$W_y \times M_{y,x} \hookrightarrow W_x$$

for all $y, x \in \mathscr{C}$, such that:

$$\dim W_x = n - |x|\,.$$

Definition 3.25. Let $\mathscr{C}$ be a flow category. A **left flow module** V of **degree** p over $\mathscr{C}$ consists of a collection of manifolds with corners $\{V_x\}_{x\in\mathscr{C}}$ together with compatible corner structure maps (embeddings):

$$M_{y,x} \times V_x \hookrightarrow V_y$$

for all $y, x \in \mathscr{C}$, such that:

$$\dim V_x = |x| - p.$$

Definition 3.26. Let $\mathscr{C}$ be a neatly embedded, normally framed flow category. A **neatly embedded, normally framed right flow module** W of degree n over $\mathscr{C}$ consists of a right flow module structure together with a choice of integer N, neat embeddings of manifolds with corners:

$$e_x : W_x \hookrightarrow \mathbb{R}^{N(n+1-|x|)} \times \mathbb{R}^{J_{n+1,x}}_{0\le}$$

and stable normal framings φ_x on the normal bundles νe_x that are compatible with the framing:

$$\partial_y \varphi_x = \varphi_y \oplus \varphi_{y,x}$$

where $\varphi_{y,x}$ is the framing on the morphism manifold $M_{y,x}$ of $\mathscr{C}$.

Similarly, we can define a neatly embedded, normally framed left flow module.

A usfeul constrution, which allow us to use the flow categories tools on flow modules is the following:

Definition 3.27. Let W be a right flow module of degree i over a flow category $\mathscr{C}$, with flow manifolds $M_{y,x}$. We define the **Cone Flow Category** $\mathrm{Cone}(W)$ as follows:

- The objects are $\mathrm{ob}(\mathrm{Cone}(W)) = \mathrm{ob}(\mathscr{C}) \sqcup \{z\}$, where z is a new formal object with index $|z| = i + 1$.

- For $x, y \in \mathscr{C}$, the morphism spaces remain identical: $(\mathrm{Cone}(W))_{y,x} = M_{y,x}$.
- For $x \in \mathscr{C}$, the morphism space from z to x is defined as the module manifold: $(\mathrm{Cone}(W))_{z,x} = W_x$. (Note that $\dim(W_x) = i - |x| = |z| - |x| - 1$, which satisfies the dimension axiom for flow categories).
- The morphism spaces from x to z are empty.

The corner structure maps

$$\coprod_{y \in \mathscr{C}} (\mathrm{Cone}(W))_{z,y} \times (\mathrm{Cone}(W))_{y,x} \hookrightarrow \partial(\mathrm{Cone}(W))_{z,x}$$

are given exactly by the module structure maps $\coprod_y W_y \times M_{y,x} \hookrightarrow \partial W_x$. If $\mathscr{C}$ and W are normally framed, these framings naturally endow the flow category $\mathrm{Cone}(W)$ with a normal framing.

Definition 3.28. Let $\mathscr{C}$ be a framed flow category, and let $n, p, r \in \mathbb{Z}$ with $r \geq 0$. Let $W_p = \{W_z\}_{|z|=p}$ be a collection of closed framed manifolds representing an element in:

$$E_1^{n,p} = \bigoplus_{|z|=p} \Omega^{\mathrm{fr}}_{n-p}.$$

We say that the collection W_p **extends to a right flow module of degree** n **over** $\mathscr{C}_{[p,p-r]}$ if there exists a neatly embedded, normally framed right flow module W of degree n over $\mathscr{C}_{[p,p-r]}$ whose restriction to the objects of degree p satisfies $W_z = (W_p)_z$ for all $z \in \mathscr{C}_p$.

Example 3.29. Let $\mathscr{C}$ be a framed flow category with objects $\{y_i\}$ and $\{x_j\}$ concentrated in only two degrees, $|y_i| = q$ and $|x_j| = p$. The morphism space:

$$M = \bigsqcup_{i,j} M_{y_i,x_j}$$

is a closed framed manifold of dimension $q - p - 1$, since there are no intermediate objects with degrees between p and q. In this case, $\mathscr{C}$ defines an element in the framed bordism group $\Omega^{\mathrm{fr}}_{q-p-1}$.

4 Chain complex of Flow Category

In this section, we establish a connection between chain complexes and framed flow categories by introducing a topologically enriched category $\mathcal{J}$, which serves as a replacement for the category of chain complexes Ch. This idea originates from the work of Cohen, Jones, and Segal [8], and is further developed in the revised version [7].

We begin by defining the pointed topologically enriched category $\mathcal{J}$, which serves as the indexing category for chain complexes in the topologically enriched setting.

Definition 4.1. We define the pointed topologically enriched category $\mathcal{J}$ as follows:

- The objects of $\mathcal{J}$ are the integers together with a distinguished object ∞:

$$\mathrm{ob}(\mathcal{J}) = \mathbb{Z} \cup \{\infty\}.$$

- The morphism spaces $\mathrm{Hom}_{\mathcal{J}}(j,i)$ are defined for $i,j \in \mathbb{Z} \cup \{\infty\}$ by:

$$\mathrm{Hom}_{\mathcal{J}}(j,i) = \begin{cases} \{\mathrm{id}, \infty\} & \text{if } i = j, \\ \{d_i, \infty\} & \text{if } j = i+1, \\ (\mathbb{R}_{0\le}^{J_{j,i}})_+ & \text{if } j > i+1, \\ \{\infty\} & \text{if } i = \infty \text{ or } j = \infty, \end{cases}$$

  where X_+ denotes the one-point compactification of a space X, and $\mathbb{R}_{0\le}$ is the half-line $[0,\infty)$. The set $J_{j,i}$ is defined by:

$$J_{j,i} = \{\ell \in \mathbb{Z} \mid j > \ell > i\}. \tag{32}$$

  The spaces $\mathrm{Hom}_{\mathcal{J}}(j,i)$ are pointed at the point ∞.

- The maps

$$J_{k,j} \sqcup J_{j,i} \hookrightarrow J_{k,i}$$

  induce a map

$$\mathbb{R}_{0\le}^{J_{k,j}} \times \mathbb{R}_{0\le}^{J_{j,i}} \hookrightarrow \mathbb{R}_{0\le}^{J_{k,i}}.$$

  By taking $(-)_+$ we obtain the composition in $\mathcal{J}$:

$$(\mathbb{R}_{0\le}^{J_{k,j}})_+ \wedge (\mathbb{R}_{0\le}^{J_{j,i}})_+ \to (\mathbb{R}_{0\le}^{J_{k,i}})_+.$$

Remark 4.2. Intuitively, the morphisms in $\mathcal{J}$ correspond to the differentials and higher compositions in a chain complex. The space $\mathbb{R}_{0\le}^{J_{j,i}}$ models the higher nullhomotopies between compositions of differentials.

Definition 4.3. Let C be a pointed topologically enriched category. We define (topological, coherent) chain complexes in C as strict functors $\mathrm{Fun}_+(\mathcal{J}, C)$.

Let Sp^O be the category of orthogonal spectra. This is a topologically enriched model category for the stable homotopy category, as developed in [17]. It is equipped with a symmetric monoidal smash product $\otimes$ that gives a Quillen equivalent model to the ∞-category of spectra. In this model, the suspension spectrum functor

$$\Sigma^\infty : (Top_+, \wedge, S^0) \to (Sp^O, \otimes, \mathbb{S})$$

is strongly symmetric monoidal and preserves coproducts.

Construction 4.4. Let $\mathscr{C}$ be a flow category with neat embedding e and normal framing φ. We define $K_{\mathscr{C}} \in \mathrm{Fun}_+(\mathcal{J}, \mathbf{Sp}^O)$ by

$$K_{\mathscr{C}}(i) = \bigoplus_{|x|=i} \mathbb{S}$$

To define the morphisms it's suffice to we will construct maps

$$\mathrm{Hom}_{\mathcal{J}}(j,i) \to \mathrm{Hom}_{\mathbf{Top}_+}(S^{N(j-i)}, S^{N(j-i)})$$

and then by taking Σ^∞_+ and desuspending we get a map

$$\mathrm{Hom}_{\mathcal{J}}(j,i) \to \mathrm{Hom}_{\mathbf{Sp}^O}(\mathbb{S},\mathbb{S})$$

we use the $(\wedge, \mathrm{Hom})$ adjunction. It suffices to provide maps:

$$g_{y,x} : \mathrm{Hom}_{\mathcal{J}}(j,i) \otimes \mathbb{S} \to \mathbb{S}$$

for each pair $y, x \in \mathscr{C}$ with indices j, i. Let y and x be as above; we define the maps

$$S^{N(|y|-|x|)} \otimes (\mathbb{R}^{J_{y,x}}_{0\leq})_+ \to Th(e_{y,x}) \to S^{N(|y|-|x|)},$$

where the first map is the collapse map of $e_{y,x}$ and the second is (the compactification of) the projection of $\varphi_{y,x}$ (see Eq. (27)). By taking Σ^∞_+ and desuspending this map, we obtain:

$$g_{y,x} : \mathbb{S} \otimes (\mathbb{R}^{J_{y,x}}_{0\leq})_+ \to Th(e_{y,x}) \to \mathbb{S}.$$

Functoriality of $K_{\mathscr{C}}$ amounts to the equation

$$\partial_y g_{z,x} = g_{y,x} \circ g_{z,y},$$

which holds due to the compatibility of the neat embedding and the framing (see Eq. (31)).

We call a functor $\mathrm{Fun}_+(\mathcal{J}, \mathbf{Top}_+)$ a **topological chain complex**. Our next goal is to start with an embedded framed flow category and produce such a topological chain complex. The idea is to use the Pontryagin-Thom collapse map (Eq. (27)) in a compatible way to "collapse" the flow category and then use $\mathcal{J}$ to encode the maps between spheres and their higher homotopies.

Construction 4.5. Let $\mathscr{C}$ be a flow category with a neat embedding e and normal framing φ. Assume that $\mathscr{C}$ has bounded index

$$b \geq |-| \geq a.$$

why we need this? how CK of CJS do this construction?

Without loss of generality, we take $a = 0$. We define a pointed topologically enriched functor $K_{\mathscr{C}} \in \mathrm{Fun}_+(\mathcal{J}, \mathbf{Top}_+)$ as follows.

First, we set

$$K_{\mathscr{C}}(i) = \bigvee_{|x|=i} S^{Nb}.$$

To define the morphisms, we use the $(\wedge, \mathrm{Hom})$ adjunction. It suffices to provide maps:

$$g_{y,x} : \mathrm{Hom}_{\mathcal{J}}(j,i) \wedge S^{Nb} \to S^{Nb}$$

for each pair $y, x \in \mathscr{C}$ with indices j, i. Let y and x be as above; we define the maps

$$S^{N(|y|-|x|)} \wedge (\mathbb{R}^{J_{y,x}}_{0\leq})_+ \to Th(e_{y,x}) \to S^{N(|y|-|x|)},$$

where the first map is the collapse map of $e_{y,x}$ and the second is (the compactification of) the projection of $\varphi_{y,x}$ (see Eq. (27)). By suspending this map, we obtain:

$$g_{y,x} : S^{Nb} \wedge (\mathbb{R}^{J_{y,x}}_{0\leq})_+ \to Th(e_{y,x}) \to S^{Nb}.$$

Functoriality of $K_{\mathscr{C}}$ amounts to the equation

$$\partial_y g_{z,x} = g_{y,x} \circ g_{z,y},$$

which holds due to the compatibility of the neat embedding and the framing (see Eq. (31)).

For a flow category $\mathscr{C}$, we define $\mathscr{C}^{[b,a]}$ to be the full subcategory generated by objects $x \in \mathscr{C}$ satisfying the index condition

$$b \geq |x| \geq a.$$

We denote by $K_{[b,a]}$ the topological chain complex of $\mathscr{C}^{[b,a]}$. Note that there is a natural transformation

$$\beta_{b+1} : K_{[b,a]} \to K_{[b+1,a]}$$

induced by suspending further to $N(b+1)$; if $i = b+1$, we take $\beta_{b+1}(i)$ to be the zero map. Similarly, we have maps

$$\alpha_{a-1} : K_{[b,a-1]} \to K_{[b,a]}.$$

Do the following def works? do we get a functor from $\mathcal{J}$?, see also the last todo

Definition 4.6. We extend Construction 4.5 to unbounded flow categories by taking the limit and colimit:

$$K_{\mathscr{C}} = \lim_a \operatorname{colim}_b K_{[b,a]}.$$

Construction 4.7. Let $\mathscr{C}$ be an embedded framed flow category bounded by b and 0. Using Construction 4.5 and post-composition with Σ^∞_+, we construct a functor $K'_{\mathscr{C}} \in \mathrm{Fun}(J, \mathbf{Sp}^O)$ with

$$K'_{\mathscr{C}}(i) = \bigoplus_{|x|=i} \mathbb{S}^{Nb}.$$

Using post-composition with $\Sigma^{-Nb} : \mathbf{Sp} \to \mathbf{Sp}$, we define the chain complex of $\mathscr{C}$ to be

$$K_{\mathscr{C}} := \Sigma^{-Nb} K'_{\mathscr{C}}$$

which has $\mathbb{S}$ in each level.

Proposition 4.8. *There is an equivalence of pointed ∞-categories $N(\mathrm{Ch}) \simeq N_{\mathbf{Top}}(\mathcal{J})$, where* Ch *is the indexing category of chain complexes (see Definition 2.2) and $N, N_{\mathbf{Top}}$ are the nerve and the topological nerve, respectively.*

Proof. The morphism spaces $\mathrm{Hom}_{\mathcal{J}}(j,i)$ are contractible when $j > i+1$ (since $(\mathbb{R}^{J_{j,i}}_{0\leq})_+$ is contractible). Furthermore, we have the following identifications:

$$\{\mathrm{id}, \infty\} \simeq S^0 \text{ for } i = j, \quad \text{and} \quad \{d_i, \infty\} \simeq S^0 \text{ for } j = i+1.$$

The equivalence follows by considering both Ch and $\mathcal{J}$ as models for the same ∞-category. □

Proposition 4.9. *In light of this equivalence, we can start with a framed flow category and obtain a coherent chain complex in spectra:*

$$\mathrm{Fun}_{\mathbf{Top}}(\mathcal{J}, \mathbf{Sp}^O) \simeq \mathrm{Fun}(N_{\mathbf{Top}}(\mathcal{J}), N_{\mathbf{Top}}(\mathbf{Sp}^O)) \simeq \mathrm{Fun}(N(\mathrm{Ch}), \mathbf{Sp}) =: \mathrm{Ch}(Sp).$$

By identifying 1-categories with their nerves, we end this section by motivating the next. Let $\mathscr{C}$ be a bounded, embedded, framed flow category and let $K \in \mathrm{Fun}(\mathrm{Ch}, \mathbf{Sp})$ be its coherent chain complex. By Section 2, we obtain a spectral sequence $(E_r^{n,p}, d_r)$ where

$$E_1^{n,p} = \pi_{n-p} K_p = \bigoplus_{|x|=p} \pi_{n-p}\mathbb{S}.$$

Identifying stable homotopy groups with framed bordism via the isomorphism

$$\pi_{n-p}\mathbb{S} \simeq \Omega^{\mathrm{fr}}_{n-p},$$

we observe that for $y, x \in \mathscr{C}$ with $|y| = p$ and $|x| = p-1$, the first differential

$$d_1 \colon \Omega^{\mathrm{fr}}_{n-p}\langle y\rangle \to \Omega^{\mathrm{fr}}_{n-p}\langle x\rangle$$

is given by the formula

$$d_1(N) = N \times M_{y,x},$$

where $M_{y,x}$ is the morphism manifold in $\mathscr{C}$ and N is a representative for its bordism class. Survival to the second page corresponds to the existence of a flow category with objects s, y, x and morphism manifolds

$$M_{y,x}, \quad N = M_{s,y}, \quad \text{and} \quad W = M_{s,x}$$

such that their boundaries satisfy the relation

$$\partial W = M_{y,x} \times N.$$

5 Models for Chain Complexes

Let $b \leq a$, to integers and let $\mathscr{C}$ be an $[b, a]$ bounded framed flow category.

The goals of this section are the following:

1. Given a right framed flow module $W : *[b] \to \mathscr{C}$, to construct a morphism $K_W : K_{*[b]} \to K_{\mathscr{C}}$ in $\mathrm{Ch}(\mathbf{Sp})$.
2. Given a left framed flow module $V : \mathscr{C} \to *[a]$, define thier composition $V \circ W : *[b] \to *[a]$, which is also a $b - a$ dimensional framed manifold, in such a way that the following diagram commutes:

$$\begin{tikzcd}
R\,\mathrm{Mod}_b(\mathscr{C}) \times L\,\mathrm{Mod}_a(\mathscr{C}) \arrow[r, "K_- \times K_-"] \arrow[d, "\circ"'] & \mathrm{Hom}_{\mathrm{Ch}(\mathbf{Sp})}(\mathbb{S}_b, K_{\mathscr{C}}) \times \mathrm{Hom}_{\mathrm{Ch}(\mathbf{Sp})}(K_{\mathscr{C}}, \mathbb{S}_a) \arrow[d, "\circ"] \\
\mathbf{Mfld}^{\mathrm{fr}}_{b-a} \arrow[r, "PT"'] & \mathrm{Hom}_{\mathrm{Ch}(\mathbf{Sp})}(\mathbb{S}_b, \mathbb{S}_a)
\end{tikzcd}$$

where PT is the Pontryagin-Thom construction and later we will justify that

$$\mathrm{Hom}_{\mathrm{Ch}(\mathbf{Sp})}(\mathbb{S}_b, \mathbb{S}_a) \simeq \mathrm{Hom}_{\mathbf{Sp}}(\Sigma^{b-a}\mathbb{S}, \mathbb{S})$$

The section layout is: In ??? we define K_V, for a module V. To do so we need to "flatten" the module into the cone flow category (see Definition 3.27), use Section **??** to obtain a chain complex and then to "unflatten" the chain complex into a morphism between chain complexes. re-wording? The main mechanism is given by modeling chain complexes using cubical diagrams. In ??? we spend some time by undersdtanding the composition

$$\mathrm{Hom}_{\mathrm{Ch}(\mathbf{Sp})}(\mathbb{S}_b, K_{\mathscr{C}}) \times \mathrm{Hom}_{\mathrm{Ch}(\mathbf{Sp})}(K_{\mathscr{C}}, \mathbb{S}_a) \to \mathrm{Hom}_{\mathrm{Ch}(\mathbf{Sp})}(\mathbb{S}_b, \mathbb{S}_a)$$

where $\mathrm{Ch}(\mathbf{Sp})$ is modeled via $\mathbf{Cat}_{\mathbf{sSet}}$ and $\mathbf{Cat}_{\mathbf{Top}}$. re-woring

In ??? we will define the Composition

$$R\,\mathrm{Mod}_b(\mathscr{C}) \times L\,\mathrm{Mod}_a(\mathscr{C}) \to \mathbf{Mfld}^{\mathrm{fr}}_{b-a}$$

and shows that the diagram commutes,

5.1 Flattening $n+1$ cubes

Definition 5.1. Let A be a finite linearly ordered set, the A-*cube* is the power set category:

$$\{1 \to 0\}^A \cong \mathcal{P}(A)^{\mathrm{op}}$$

Let $n \in \mathbb{N}$ (including 0). For $A = \{1, \dots, n\}$ (or $\varnothing$ if $n = 0$), we write $\{1 \to 0\}^n = \{1 \to 0\}^A$.

Note. For brevity we will use [1] along this section, and at the end we will "opposite" it so our cubes (and chains) will be decreasing and not increasing.

Definition 5.2. Let A be a finite linearly ordered set, let $a \in A$, define $s_a \subseteq A$ to be the prefix subset

$$s_a := \{b \in A \mid b \leq a\}.$$

The spine of A is the full subcategory of $[1]^A$ spanned by

$$\mathrm{Spine}(A) := \{\varnothing\} \cup \{s_a \mid a \in A\}$$

Definition 5.3. Let $\mathcal{C}$ be a pointed ∞-category and let A be a finite linearly ordered set. We define the category of spine cubes (or cubical chain complexes)

$$\mathrm{Cu}_A(\mathcal{C}) := \mathrm{Fun}^{\mathrm{Spine}}([1]^A, \mathcal{C}) := \{F \in \mathrm{Fun}([1]^A, \mathcal{C}) \mid F(Q) \simeq * \text{ for all } Q \notin \mathrm{Spine}(A)\}$$

Let A, B be finite linearly ordered sets, then

$$[1]^{A \amalg B} \simeq [1]^A \times [1]^B,$$

hence

$$\mathrm{Fun}([1]^{A \amalg B}, \mathcal{C}) \simeq \mathrm{Fun}([1]^A \times [1]^B, \mathcal{C}) \simeq \mathrm{Fun}([1]^A, \mathrm{Fun}([1]^B, \mathcal{C})).$$

By taking $A = \{a\}$, we obtain

$$\mathrm{Fun}([1]^{B \amalg \{a\}}, \mathcal{C}) \simeq \mathrm{Fun}([1]^B \times [1]^{\{a\}}, \mathcal{C}) \simeq \mathrm{Fun}([1], \mathrm{Fun}([1]^B, \mathcal{C})).$$

We think of this equivalence as "flattening" the morphism of cubes (RHS) into a longer cube (the LHS).

Passing to the spine cubes we need to be careful: Let $B = \{1, \dots, n\}$ be a linearly ordered set, and let $a = n + 1$ be an element larger than all elements of B. We have that the spine of $B \amalg \{a\}$ inside $[1]^{B \amalg \{a\}} \simeq [1]^B \times [1]$ consists of the elements:

$$(0, \dots, 0, 0), \quad (1, 0, \dots, 0, 0), \quad \dots, \quad (1, \dots, 1, 0), \quad (1, \dots, 1, 1)$$

which can be written as:

$$(\mathrm{Spine}(B) \times \{0\}) \cup \{(B, 1)\} \subset [1]^B \times [1].$$

We get that for the cubical chain complexes we have:

$$\begin{aligned} \mathrm{Cu}_{B \amalg \{a\}}(\mathcal{C}) &\simeq \mathrm{Fun}^{\mathrm{Spine}}([1]^{B \amalg \{a\}}, \mathcal{C}) \\ &\simeq \left\{ f \colon [1] \to \mathrm{Fun}([1]^B, \mathcal{C}) \;\middle|\; \begin{array}{ll} f_0(Q) \simeq * & \text{for all } Q \notin \mathrm{Spine}(B), \\ f_1(Q) \simeq * & \text{for all } Q \neq B \end{array} \right\} \end{aligned} \tag{33}$$

where f_0, f_1 are the source and the target of the morphism f.

Before we move on, let us compare the coherent chain complexes defined via Ch to the cubical model.

Let $\mathcal{C}$ be a pointed ∞-category. For $F \in \mathrm{Fun}_*([1]^A, \mathcal{C})$, the condition that F is a spine cube is equivalent to saying that $F(Q) \simeq *$ for all $Q \notin \mathrm{Spine}(A)$. We can define a localized category Cu_A by starting with $[1]^A$, adding a zero object $*$, and formally adjoining morphisms $* \to Q$ for all $Q \notin \mathrm{Spine}(A)$. Inverting these morphisms yields:

$$\mathrm{Cu}_A := \big([1]^A \amalg \{*\}\big)\,[\{* \to Q \mid Q \notin \mathrm{Spine}(A)\}^{-1}]$$

This localization is constructed precisely so that for any pointed ∞-category $\mathcal{C}$, we have an equivalence of functor categories:

$$\mathrm{Fun}_*^{\mathrm{Spine}}([1]^A, \mathcal{C}) \simeq \mathrm{Fun}_*(\mathrm{Cu}_A, \mathcal{C}).$$

Under this equivalence, the map that sends "off-spine" elements to $*$ in the definition of Ch translates to the fact that Cu_A is equivalent to the corresponding subcategory of Ch. We formalize this as follows:

Claim 5.4. For any integers $b \le a$, let $A = \{b, \dots, a\}$. We have an equivalence of categories:

$$\mathrm{Cu}_{[b,a]} := \mathrm{Cu}_{\{b,\dots,a\}} \simeq \mathrm{Ch}_{[b,a]}$$

where $\mathrm{Ch}_{[b,a]}$ is the full subcategory of Ch spanned by the objects $\{b, \dots, a\} \cup \{*\}$.

Note that the restriction functor defined in Section 2 is compatible with this equivalence: for any subset $A \subseteq \{1, \dots, n\}$, the restriction functor on chain complexes $\mathrm{Ch}_n(\mathcal{C}) \to \mathrm{Ch}_A(\mathcal{C})$ corresponds to a restriction functor

$$\mathrm{Cu}_n(\mathcal{C}) \to \mathrm{Cu}_A(\mathcal{C}).$$

This restriction allows us to cleanly phrase the relationship between an $(n+1)$-dimensional spine cube and a morphism of n-dimensional spine cubes:

Lemma 5.5. *Let $A = \{1, \dots, n\}$ and let $B = A \amalg \{n+1\} = \{1, \dots, n+1\}$. By Eq. (33), an object $K \in \mathrm{Cu}_B(\mathcal{C})$ (which is an $(n+1)$-dimensional cubical chain complex) corresponds precisely to a morphism $f\colon L \to M$ in $\mathrm{Fun}([1]^A, \mathcal{C})$ characterized by the following restrictions:*

1. *The restriction of K to the 0-face of the $(n+1)$-th direction gives $L = f_0$, which is an n-dimensional cubical chain complex in $\mathrm{Cu}_A(\mathcal{C})$.*
2. *The restriction of K to the 1-face of the $(n+1)$-th direction gives $M = f_1$, which must be concentrated at the final object of $[1]^A$ (i.e., at A).*

Remark 5.6. Symmetrically, if we instead consider an index set of the form $\{1\} \amalg B'$ where $B' = \{2, \dots, n+1\}$ (i.e., prepending an element), the spine equivalence identifies an $(n+1)$-dimensional cubical chain complex $K \in \mathrm{Cu}_{\{1\}\amalg B'}(\mathcal{C})$ with a morphism $f\colon L \to M$ in $\mathrm{Fun}([1]^{B'}, \mathcal{C})$. Under this identification, the source $L = f_0$ is concentrated at the initial object $\varnothing$, while the target $M = f_1$ is a general n-dimensional cubical chain complex in $\mathrm{Cu}_{B'}(\mathcal{C})$.

Now we can "flatten" a flow module into a flow category, realize it into a cubical chain complex, and then "unflatten"

6 Differential in a flow category

By passing the definition of the differential (see Proposition 2.32) through the topological model of chain complexes, we can start with a flow category and get a flow categorical formula for the differential. Using this description we will extract an sufficient and necessary condition for the survival of an element.

Let W, V be right (resp. left) framed flow modules, of degree i (resp. j) over a framed flow category $\mathscr{C}$. Note that $V \circ W$ is a right framed module of degree i over the point flow category $*[j]*$ witch is the data of a single $i-j$ dimensional framed manifold, which we also call $V \circ W$. On the other hand W, V give rise to morphisms K_W, K_V (see **??**) of coherent chain complexes such that the composition is a map of coherent chain complexes:

$$K_V \circ K_W : \mathbb{S}_i \to \mathbb{S}_j$$

Using lemma Lemma 2.31 we get that $K_V \circ K_W$ is equivalent to a map of spectra:

$$\mathbb{S}^{i-j} \to \mathbb{S}$$

which, by Thom Pontryagin is a cobordism class in Ω^{fr}_{i-j}, our goal is to show that we actually obtained the cobordism class of $V \circ W$.

Note: at the end $V \circ W$ constructed only from the subcategory $\mathscr{C}_{i,j}$, so from this point we assume for simplicity that $\mathscr{C} = \mathscr{C}_{i,j}$.

Let W be a flow bimodule over a flow category $\mathscr{C}$. We have the cone flow category $\text{Cone}(W)$, and its coherent chain complex $K_{\text{Cone}(W)}$. Note that:

$$(\text{Cone}(W))_{i,j} = \mathscr{C}$$

in sense of objects and index and mortphism manifolds. Realizing to get that:

more that into a defin- tion from itself?

$$(K_{\text{Cone}(W)})_{\geq i} \simeq (K_{\text{Cone}(W)})_{i,j} \simeq K_{(\text{Cone}(W))_{i,j}} \simeq K_{\mathscr{C}}$$

Also note that $\text{Cone}(W)_{i+1}$ consist of one object, hence

$$(K_{\text{Cone}(W)})_{i+1} \simeq \mathbb{S}.$$

Applying **??** to the chain complex $K_{\text{Cone}(W)}$ restricted to $[i, j]$ we get a morphism, of the form:

$$\mathbb{S}_i \to (K_{\text{Cone}(W)})_{i,j} \simeq K_{\mathscr{C}}.$$

which we think of as the "flattening" of the cone flow category.

By abuse of notation we denote:

$$K_W := K_{\text{Cone}(W)}$$

and also the morphism which obtain from this chain complex:

$$K_W : \mathbb{S}_i \to (K_{\text{Cone}(W)})_{i,j} \simeq K_{\mathscr{C}}.$$

Similarly for a left framed flow module V of degree j over $\mathscr{C}$ we get a freamed flow category K_V and a morphism:

$$K_V : K_{\mathscr{C}} \to \mathbb{S}_j.$$

Note taht by the deinfiton of the cone flow category we have that the morphism manifolds of $\mathrm{Cone}(W)$ are the manifolds W_x and $M_{y,x}$, hence the chain comlex K_W is obtained by the Thom Pontryagin bordism collapces maps of (a framed neat embbeding of) W_x and $M_{y,x}$.

Using ?? we get the composition of K_W, K_V, as a map:

$$S^{i-j} \to \mathrm{Hom}_{\mathbf{Sp}^O}(\mathbb{S}, \mathbb{S}),$$

is give by

Note that:

$$(\mathrm{Cone}(W))_{i,j} = \mathscr{C}$$

in sense of objects and index and mortphism manifolds. Realizing to get that:

more that into a defin- tion from itself?

$$(K_{\mathrm{Cone}(W)})_{\geq i} \simeq (K_{\mathrm{Cone}(W)})_{i,j} \simeq K_{(\mathrm{Cone}(W))_{i,j}} \simeq K_{\mathscr{C}}$$

Similarly we get a map

$$(K_{\mathrm{Cone}(W)})_{i+1} \simeq K_{(\mathrm{Cone}(W))_{i+1}} \simeq K_{*[i+1]} \simeq \mathbb{S}_{i+1}$$

By using the adjoinction of the truncation from above, see Eq. (15), we get an inclusion

$$K_{\mathscr{C}} \simeq (K_{\mathrm{Cone}(W)})_{\geq i} \hookrightarrow K_{\mathrm{Cone}(W)}.$$

Cofibers in functor categories are calculated pointwise, hence :

$$K_{\mathrm{Cone}(W)}/K_{\mathscr{C}} \simeq (K_{\mathrm{Cone}(W)})_{i+1} = \mathbb{S}_{i+1}$$

and a cofiber sequence

$$K_{\mathscr{C}} \to K_{\mathrm{Cone}(W)} \to \mathbb{S}_{i+1}$$

which, by rolling, give a conenctive morphism:

$$\mathbb{S}_{i+1} \to \Sigma K_{\mathscr{C}}.$$

Using Lemma 2.31 we get a map:

$$\mathbb{S}_i \to K_{\mathscr{C}}.$$

We start be intruducing the following abuse of notaitons: We denote:

$$K_W \coloneqq K_{\mathrm{Cone}(W)}$$

and also the induces morphims is denoted the same:

$$K_W : \mathbb{S}_i \to K_{\mathscr{C}}.$$

Similarly a left flow module V of degree j on $\mathscr{C}$ is induces a map:

$$K_V : K_{\mathscr{C}} \to \mathbb{S}_j.$$

A short preliminariy to this section will be to define the realization of a flow module.

PLACE HOLDER: DEF OF Cone flow category. replace

Definition 6.1. Let $b, a \in \mathbb{Z}$. Let $\mathscr{C}$ and $\mathscr{D}$ be framed flow categories bounded by $[b+1, a]$ and $[b, a-1]$ respectively. We say that $\mathscr{C}$ and $\mathscr{D}$ are **composable** if

$$\mathscr{C}_{[b,a]} = \mathscr{D}_{[b,a]}$$

where $\mathscr{C}_{[b,a]}$ is the full subcategory generated by $c \in \mathscr{C}$ such that $b \geq |x| \geq a$.

Similarly two chain complexes K, L (with same bounds) are composable if K, L are composable as morphisms (see **??**).

Note that the target of K is $K_{[b,a]}$ and the source of L is $L_{[b,a]}$ we deduce that for any composable categories $\mathscr{C}, \mathscr{D}$, we get that $K_{\mathscr{C}}, K_{\mathscr{D}}$ are composable chain complexes.

Our goal is to define $\mathscr{D} \circ \mathscr{C}$ to be a flow category such that

$$K_{\mathscr{D}\circ\mathscr{C}} \simeq K_{\mathscr{D}} \circ K_{\mathscr{C}}.$$

Let W, V be right and left flow bimodules over a flow category, $*[i] \xrightarrow{W} \mathscr{C} \xrightarrow{V} *[j]$. Our goal is to define the flow module $W \circ V :\in [*[i], *[j]]$. from here

Claim 6.2. Let W, V be two composable framed flow bimodules over a framed flow category $\mathscr{C}$. Then

$$\bigcup_{i \geq s \geq j} W_s \times V_s$$

is a closed framed manifold with dimension $i - j$.

The above claim will follow from the general following "flattening" argument.

Lemma 6.3. *Let A be a finite set, $N \in \mathbb{N}$ and let*

$$D : \{1 \to 0\}^A \setminus \underline{0} \to \mathbf{Top}$$

be a diagram such that for each $a \in A$ the restriction

$$\{1 \to 0\}^{A \setminus \{a\}} \to \{1 \to 0\}^A \to \mathbf{Top}$$

is $|A| - 1$ dimensional $\langle A \setminus \{a\} \rangle$ manifold. Then the topological space

$$X := \operatorname*{colim}_{\{1\to 0\}^A \setminus \underline{0}} D$$

inherits a structure of a closed $|A| - 1$ manifold. Moreover, if $D(a)$ are framed, then X inherits a framing.

We refer to such diagrams as "punctured cubes".

Proof. The idea is to use the following homeomorphism:

$$\mathbb{R}^2_{0\leq} \simeq \mathbb{R}_{0\leq} \times \mathbb{R}$$

which, in polar coordinates, is given by

$$(r, \theta) \mapsto (r, 2\theta).$$

Inductively we get that for any finite set A and $a \in A$ we have

$$\mathbb{R}^A_{0\leq} \simeq H \times \mathbb{R}^{A\setminus\{a\}}$$

which induces the homeomorphism on the boundary

$$\partial\mathbb{R}^A_{0\leq} \simeq \mathbb{R}^{A\setminus\{a\}}, \tag{34}$$

see Definition 3.1.

The fact that the restrictions of D are $\langle A \setminus \{a\}\rangle$ manifolds give them the local model of

$$\mathbb{R}^{A\setminus\{a\}}_{0\leq}$$

hence, for $B \subset A, a \in B$, locally $D(B)$ is given by

$$\mathbb{R}^{A\setminus\{a\}}_{0\leq}(B) = \partial_a\mathbb{R}^A_{0\leq}(B)$$

where $\mathbb{R}^A_{0\leq}$ is the diagram of the manifolds with corners see Construction 3.11 and we view $\partial_a\mathbb{R}^A_{0\leq}$ as a sub-diagram of $\mathbb{R}^A_{0\leq}$. Under this identification we have that

$$\partial_a\mathbb{R}^A_{0\leq}(B) = \mathbb{R}^A_{0\leq}(B)$$

where $D(B)$ is locally homeomorphic to $\mathbb{R}^A_{0\leq}(B)$.

By taking colimit:

$$\operatorname*{colim}_{\{1\to 0\}^A\setminus \underline{0}} \mathbb{R}^A_{0\leq} = \partial(\mathbb{R}^A_{0\leq})$$

we get that any point of X have an neighborhood homeomorphic to an open set of $\partial\mathbb{R}^A_{0\leq}$, to get a manifold structure on X we use Eq. (34) and get that X is locally given by

$$\partial(\mathbb{R}^A_{0\leq}) \simeq \mathbb{R}^{A\setminus\{a\}} \simeq \mathbb{R}^{|A|-1}.$$

Assume further that $D(a)$ are embedded and normally framed manifolds with framing φ_a. The natural transformations φ_a

$$\nu D(a) \simeq D(a) \times \mathbb{R}^N$$

(where $N \in \mathbb{N}$) assemble into a neutral transformation

$$\nu D \simeq D \times \mathbb{R}^N$$

where the left is the diagram $\nu D(B)$ and the right is $D(B) \times \mathbb{R}^N$. By taking colimit we get a normal framing

$$\nu X \simeq X \times \mathbb{R}^N.$$

□

another approach is to factor the structure maps $D(B) \to BO(?)$ through $*$, not sure if this is too homotopical

Example 6.4. Take $\mathscr{C}$ to be a frmaed flow category with two objects with indices $1, 0$ and let W, V be right and left bimodules over $\mathscr{C}$ with degrees $1, 0$ respectively. The data of $\mathscr{C}$ is a manifold $M_{1,0}$, and W, V amounts to a closed 0-dimensional manifolds W_1, V_0 and frmaed 1-dimensional manifolds W_0, V_1 with boundary

$$\partial W_0 \simeq W_1 \times M_{1,0}, \quad \partial V_1 \simeq M_{1,0} \times V_0.$$

The construction is given by gluing

$$W \circ V \simeq W_0 \times V_1 \cup V_0 \times W_1$$

along $W_1 \times M_{1,0} \times V_0$.

This example recovers the secondary structure on bordism rings, introduced by Alexander in [2]. We can think of $W \circ V$ as the *Toda brackets* or *Massey product* of the 0-dimensional manifolds $W_1, M_{1,0}, V_0$.

verity in Alexander which one is the right

Definition 6.5. Let W, V be right (left) modules over a flow category $\mathscr{C}$ with degree i (resp j), with flow manifolds $W_{-}, M_{-,-}, V_{-}$. We define

$$D\colon \{1 \to 0\}^{\{i,\dots,j\}} \setminus \underline{0} \to \mathbf{Top}$$

in the following way: for each $B = \{b_k, \dots b_1\} \subseteq \{i, \dots, j\}$

$$D(b_k, \dots b_1) = W_{b_k} \times M_{b_k, b_{k-1}} \times \cdots \times M_{b_2, b_1} \times V_{b_1}.$$

We write

$$W \circ V := \bigcup_{i \leq p \leq j} W_p \times V_p := \operatorname{colim} D.$$

Note that $W \circ V : *[i] \to *[j]$, hence consist of only one manifold, so this notation consistent with the flow modules notation of the paper.

Claim 6.6. Let D be a punctured diagram as in Lemma 6.3, then we have that

$$(\operatorname{colim} D(B))_+ \simeq \operatorname{colim}(D(B)_+)$$

Proof. Note that

$$\operatorname{colim} D(B)_+ = \operatorname{colim}(D(B) \cup \{\infty_B\}) \simeq (\operatorname{colim} D(B)) \cup \{\infty_A\}$$

where that last equivalence is the identification of all the points with the point of $D(A)$. This resulting a compact space, which is differ from $\operatorname{colim} D$ by one point, hence we get that

$$\operatorname{colim} D(B)_+ \simeq (\operatorname{colim} D)_+$$

□

Lemma 6.7. *Let $\mathscr{C}, \mathscr{D}$ be two composable framed flow categories, then We have that*

$$K_{\mathscr{D} \circ \mathscr{C}} \simeq K_{\mathscr{D}} \circ K_{\mathscr{C}}$$

Proof. First we remind that $K_{\mathscr{C}}$ is the chain complexes with maps

$$f_{z,y} : \mathbb{R}^{J_{z,y}}_{0 \leq} \wedge S^{N(b+1-a)} \to S^{N(b+1-a)}$$

which constructed by collapsing $M_{z,y}$. Similarly the chain complex $K_{\mathscr{D}}$ have the maps $g_{y,x}$ which constructed by collapsing $N_{y,x}$.

Using the topologically enriched model, by **??** our goal is to show that

$$c(\bigcup_{b\geq p\geq a} M_{b+1,p}\times N_{p,a-1}) = \bigcup_{b\geq p\geq a} f_{b+1,p}\wedge g_{p,a-1}$$

where c is the Pontrjagin-Thom collapse map.

Let D be a punctured cube diagram as above. Let $X \coloneqq \operatorname{colim} D$ be the colimit manifold. We denote

$$A = J_{b+1,a-1} = \{b,\dots a\}$$

and note that $|A| = b-a+1$. We remind that the Pontryagin Thom map of X is constructed in the following way: for each $B\subseteq \{b,\dots a\}$ we have a neat embedding

$$e(B)\colon D(B)\hookrightarrow \mathbb{R}^A_{0\leq}(B)\times\mathbb{R}^{N(b-a+2)}$$

by taking colimit we get the assembly map

$$X = \operatorname*{colim}_{\{1\to 0\}^A\setminus \underline{0}} D \hookrightarrow \operatorname*{colim}_{\{1\to 0\}^A\setminus \underline{0}} \mathbb{R}^A_{0\leq}\times\mathbb{R}^{N(b-a+2)} \simeq \mathbb{R}^{|A|-1}\times\mathbb{R}^{N(b-a+2)}.$$

We get an open embedding of the normal bundle

$$e\colon \nu X\hookrightarrow \mathbb{R}^{|A|-1}\times\mathbb{R}^{N(b-a+2)}$$

which is the colimit of the maps

$$e(B)\colon \nu D(B)\hookrightarrow \mathbb{R}^A_{0\leq}(B)\times\mathbb{R}^{N(b-a+2)}.$$

On the other hand we have the normal framing, compost with the projection

$$\varphi(B)\colon \nu D(B)\simeq D(B)\times\mathbb{R}^{N(b-a+2)}\to\mathbb{R}^{N(b-a+2)}$$

which assemble into the map

$$\operatorname{colim}\varphi\colon \nu X\to\mathbb{R}^{N(b-a+2)}$$

The Thom map (see Eq. (27)) of X is given by

$$c_X = \varphi_+\circ e_+$$

where e_+ is given by collapsing the complement of νX. We can re-write this map as

$$(\operatorname{colim}\varphi(B))_+\circ\operatorname{colim}(e(B))_+$$

and using Claim 6.6 we get that

$$(\operatorname{colim}\varphi(B)_+)\circ\operatorname{colim}(e(B)_+) = \operatorname{colim}(\varphi(B)_+\circ e(B)_+)$$

which is the colimit over the diagram of taking point-wise Thom maps of $D(B)$. To finish the proof we need to show that the Thom maps of

$$D(\{p\}) = M_{b+1,p}\times N_{p,a-1}$$

are homotopic to $f_{b+1,p}\wedge g_{p,a-1}$. This follows from the functoriality of $(-)_+$ and by isotoping the tubular neighborhoods

$$\nu(M_{b+1,p}\times N_{p,a-1}) \text{ and } \nu M_{b+1,p}\times\nu N_{p,a-1}.$$

□

Theorem 6.8. *Let $\mathscr{C}$ be a flow category with a neat embedding and normal framing. Let $K_{\mathscr{C}}$ be the chain complex obtained by collapsing the morphism manifolds of $\mathscr{C}$, and denote by $E_r^{n,p}$ the spectral sequence of $K_{\mathscr{C}}$. We have that:*

$$E_1^{n,p} = \bigoplus_{|x|=p} \Omega^{\mathrm{fr}}_{n-p}.$$

A collection of closed framed manifolds $L = \{L_z\}_{|z|=p}$ representing an element in $E_1^{n,p}$ survives to the $(r+1)$-th page if and only if L extends to a right flow module W of degree n over $\mathscr{C}_{[p,p-r]}$ (see Definition 3.28).

If L survives to the $(r+1)$-th page, then the differential $d_{r+1}(L)$ is given by the bordism class of the Toda manifold $W \circ \mathcal{M}_{-,p-r-1}$ (the geometric tensor product of the right module W and the left module $\mathcal{M}_{-,p-r-1}$), which is represented by the closed framed manifold:

$$\bigcup_{p \geq q \geq p-r} W_q \times M_{q,p-r-1}$$

where the W_q are the manifolds in the flow module W, and the $M_{q,p-r-1}$ are the morphism manifolds of $\mathscr{C}$.

Proof. On the one hand, assume that the collection L extends to a right flow module W over $\mathscr{C}_{[p,p-r]}$ of degree n. Using Construction 4.5, we construct a map of coherent chain complexes (or a $\mathscr{J}$-module map) corresponding to W:

$$K_W : K_{S_n} \to K_{\mathscr{C}_{[p,p-r]}}$$

which extends the map representing L on the degree p objects. By the homological criterion established in Proposition 2.24, this map of chain complexes guarantees that the element L survives to the $(r+1)$-th page of the spectral sequence.

For the differential, let W be the right flow module of degree n over $\mathscr{C}_{[p,p-r]}$ extending L. By Proposition 2.32, the differential $d_{r+1}(L)$ is given by the composition of the chain maps:

$$K_{\mathcal{M}_{-,p-r-1}} \circ K_W : K_{S_n} \to K_{p-r-1}$$

where the first map is the left flow module $\mathcal{M}_{-,p-r-1}$ viewed as a chain map. Under the equivalence between compositions of chain complexes and compositions of modules, this corresponds to the Toda manifold $W \circ \mathcal{M}_{-,p-r-1}$, which is represented by the closed framed manifold:

$$\bigcup_{p \geq q \geq p-r} W_q \times M_{q,p-r-1}$$

where the W_q are the manifolds of the module W and $M_{q,p-r-1}$ are the morphism manifolds of $\mathscr{C}$.

The remaining part is to show that if $L \in E_1^{n,p}$ survives to the $(r+2)$-th page, then the collection L extends to a right flow module over $\mathscr{C}_{[p,p-r-1]}$. By the definition of survival, if L survives then $d_{r+1}(L) = 0$ in bordism. Thus, the Toda manifold $W \circ \mathcal{M}_{-,p-r-1}$ is null-bordant, meaning there exists a framed manifold with corners W_{p-r-1} of dimension $n-p+r$ such that:

$$\partial W_{p-r-1} = W \circ \mathcal{M}_{-,p-r-1} = \bigcup_{p \geq q \geq p-r} W_q \times M_{q,p-r-1}.$$

By adding the manifold W_{p-r-1} to the collection $\{W_q\}_{q \in \mathscr{C}_{[p,p-r]}}$, we obtain a right flow module over $\mathscr{C}_{[p,p-r-1]}$ of degree n, as required. □

To do it carefully, we need to find a manifold with corners structure on W_{p-r-1}, which is given by

7 Tangential structures and change of rings

In this section we point out how to generalized Theorem 6.8 to a case where we replace the framing structure on the flow category $\mathscr{C}$ by other tangential structures. As a result we get a spectral sequence that calculates the bordism homology $\Omega^{\mathcal{X}}_*(|\mathscr{C}|)$.

To start with, we will generalize the definition of normal framing.

Definition 7.1. A tangential structure is a collection of fibrations $\mathcal{X}(n) \to BO(n)$, with maps $\mathcal{X}(n) \to \mathcal{X}(n+1)$ such that

$$\begin{tikzcd} \mathcal{X}(n) \arrow[r] \arrow[d] & \mathcal{X}(n+1) \arrow[d] \\ BO(n) \arrow[r] & BO(n+1) \end{tikzcd}$$

Definition 7.2. Let $M \hookrightarrow \mathbb{R}^k_{0\le} \times \mathbb{R}^N$ be an embedded m dimensional $\langle k \rangle$ manifold. A **normal $\mathcal{X}$ structure on** M is a choice of a lift

$$\begin{tikzcd} & \underline{\mathcal{X}(N)} \arrow[d] \\ M \arrow[ur, dashed, "\nu\tilde{M}"] \arrow[r, "\nu M"'] & \underline{BO(N+k-m)} \end{tikzcd} \tag{35}$$

where the diagram takes place in the category $\langle k \rangle$ spaces, which is a functorial $\{1 \to 0\}^k$ family of lifts

$$\begin{tikzcd} & \mathcal{X}(N) \arrow[d] \\ M(A) \arrow[ur, dashed] \arrow[r, "\nu M(A)"'] & BO(N+k-m) \end{tikzcd} \tag{36}$$

for each $A \subseteq \{1 \to 0\}^k$.

Definition 7.3. We say that an $\mathcal{X}$ structure is **multiplicative** if it is equipped with structure maps

$$\mathcal{X}(n) \times \mathcal{X}(m) \to \mathcal{X}(n+m)$$

that commutes with the maps

$$BO(n) \times BO(m) \to BO(n+m)$$

Note that if $\mathcal{X}$ is multiplicative then a choice of $\mathcal{X}$ structure on two embedded manifolds M, L induces a $\mathcal{X}$ structure on $M \times L$ in the following way

$$\begin{tikzcd} & \underline{\mathcal{X}(d) \times \mathcal{X}(d')} \arrow[r] \arrow[d] & \underline{\mathcal{X}(d+d')} \arrow[d] \\ L \times M \arrow[ur, dashed] \arrow[r] & \underline{BO(d) \times BO(d')} \arrow[r] & \underline{BO(d+d')} \end{tikzcd}$$

where d and d' are the codimensions of L and M, respectively.

For an tangential structure $\mathcal{X}$ we denote by ζ_n the universal real vector bundle over $X(n)$, which is (by abuse of notation) the pullback of the universal real vector bundle $\zeta_n \to BO(n)$. Let M be an $\mathcal{X}$ structure manifold (for simplicity - a closed manifold) and denote by d the codimension of M in the ambient space, we obtain the following diagram:

$$\begin{tikzcd} \nu M \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \zeta_n \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \zeta_n \arrow[d] \\ M \arrow[r, "\tilde{\nu M}"] & X(d) \arrow[r] & BO(d) \end{tikzcd}$$

where the right square is pullback by definition and the same for the whole rectangle. We denote

$$\varphi_M \colon \nu M \to \zeta_n \tag{37}$$

Note that although we omitted $\nu \tilde{M}$ from the notation, a different choices of $\nu \tilde{M}$ defines different map φ_M. Note that if M is $\langle k \rangle$ manifold then φ_M is a map of $\langle k \rangle$ vector bundles.

Given a multiplicative structure we can define an $\mathcal{X}$ structured flow category (compare to Definition 3.23).

Definition 7.4. Let $\mathscr{C}$ be an embedded flow category, an $\mathcal{X}$ structure on $\mathscr{C}$ is a choice of $\mathcal{X}$ structure on each embedded morphism manifold $M_{y,x}$ such that

$$\partial_y \varphi_{z,x} = \varphi_{z,y} \oplus \varphi_{y,x}$$

where $\varphi_{y,x}$ is the isomorphism of $\langle J_{y,x} \rangle$ bundles arise from the $\mathcal{X}$ structure on $M_{y,x}$, see Eq. (37).

Similarly to Construction 4.5, using Pontryagin Thom we get a coherent chain complex in $\mathrm{Mod}_{M\mathcal{X}}$ where $M\mathcal{X}$ is the Thom spectrum of $\mathcal{X}$.

Construction 7.5. Let $\mathscr{C}$ be a bounded $\mathcal{X}$ flow category. For each $x, y \in \mathscr{C}$ we define a map

$$g_{y,x} : S^{N(|y|-|x|)} \wedge (\mathbb{R}^{J_{y,x}}_{0\leq})_+ \to Th(\nu M_{y,x}) \to Th(\zeta_{N(|y|-|x|)}).$$

where the first map is the collapse map of the embedding $e_{y,x}$ (see Definition 3.4) and the second map is $Th(\varphi_{y,x})$. The coherence of being $\mathcal{X}$ flow categories amounts to

$$g_{z,y} \wedge g_{y,x} = \partial_y g_{z,x}$$

as maps between diagrams in **Top**.

By taking $\Sigma^{N(|y|-|x|)} \Sigma^\infty_+ g_{y,x}$ we get a collection of maps to the Thom spectrum of $\mathcal{X}$

$$\mathbb{S} \otimes (\mathbb{R}^{J_{y,x}}_{0\leq})_+ \to M\mathcal{X}.$$

which is equivalent to a map

$$(\mathbb{R}^{J_{y,x}}_{0\leq})_+ \to \mathrm{Hom}_{\mathbf{Sp}^O}(\mathbb{S}, M\mathcal{X}),$$

the last step is to use the adjunction between $\mathbb{S} - \mathrm{Mod}$ and $M\mathcal{X} - \mathrm{Mod}$ to obtain a map

I am confuse, we can move between strict functors of topologically enriched categories to ∞ functors in **Sp**, but can we do

$$\bar{g}_{y,x} : (\mathbb{R}^{J_{y,x}}_{0\leq})_+ \to \mathrm{Hom}_{M\mathcal{X}}(M\mathcal{X}, M\mathcal{X}).$$

To get that the collection of these maps define a functor

$$\mathcal{J} \to \mathrm{Mod}-\mathfrak{X}$$

we use the following commutative diagram:

$$\begin{tikzcd}
\mathrm{Hom}_{\mathbf{Top}}(S^d,\zeta_d)\times \mathrm{Hom}_{\mathbf{Top}}(S^{d'},\zeta_{d'}) \arrow[rr, "\wedge"] \arrow[d, "(\Sigma^{-d}\Sigma^\infty_+,\Sigma^{-d'}\Sigma^\infty_+)"'] & & \mathrm{Hom}_{\mathbf{Top}}(S^{d+d'},\zeta_{d+d'}) \arrow[d, "\Sigma^{-(d+d')}\Sigma^\infty_+"] \\
\mathrm{Hom}_{\mathbf{Sp}}(\mathbb{S},M\mathfrak{X})\times \mathrm{Hom}_{\mathbf{Sp}}(\mathbb{S},M\mathfrak{X}) \arrow[d, "\mathrm{Mod}"'] & & \mathrm{Hom}_{\mathbf{Sp}}(\mathbb{S},M\mathfrak{X}) \arrow[d, "\mathrm{Mod}"] \\
\mathrm{Hom}_{M\mathfrak{X}}(M\mathfrak{X},M\mathfrak{X})\times \mathrm{Hom}_{M\mathfrak{X}}(M\mathfrak{X},M\mathfrak{X}) \arrow[rr, "\circ"'] & & \mathrm{Hom}_{M\mathfrak{X}}(M\mathfrak{X},M\mathfrak{X})
\end{tikzcd}$$

where Mod denote the $\mathbb{S}-\mathrm{Mod},\mathfrak{X}-\mathrm{Mod}$ adjunction. We obtain that

$$\bar{g}_{z,y}\otimes \bar{g}_{y,x} = \partial_y \bar{g}_{z,x}$$

hence we got a (topologically enriched) coherent chain complex $\mathrm{Fun}(\mathcal{J},\mathrm{Mod}-\mathfrak{X})$.

HEre is a question: Let $E \in \mathbf{Sp}$... do we have ... category of E modules and adjoinc- tion?

The definition of a right flow module (Definition 3.24) is modified by replacing framed flow modules by $\mathfrak{X}$ structured flow modules and Ω^{fr} by $\Omega^{\mathfrak{X}}$.

Similarly, we define when two $\mathfrak{X}$ flow categories are composable. To define the composition category we modify Definition 6.5 we get a boundary diagram

$$D\colon \{1\to 0\}^{\{b,\dots a\}}\setminus \underline{0} \to \mathbf{Top}$$

such that $D(\{a\})$ are $\mathfrak{X}$ manifolds with $\langle A\setminus\{a\}\rangle$ corner structure. he lifts of the structure map $\nu D(B)$ assemble into a lift of the structure map $\nu\operatorname{colim} D$ hence induces a $\mathfrak{X}$ structure on the colimit. We get that

Corollary 7.6. *Let $\mathscr{C},\mathscr{D}$ be two composable flow categories bounded by $b+1, a-1$, with $\mathfrak{X}$ structure and flow manifolds written as $M_{-,-}, N_{-,-}$ then*

$$\bigcup_{b\le p\le a} M_{b+1,p}\times N_{p,a-1} \coloneqq \operatorname{colim} D$$

is an $\mathfrak{X}$ manifold (compare to Definition 6.5). Where D is defined in the following way: for each $B=\{i_k,\dots i_1\}\subseteq\{b,\dots a\}$

$$D(i_k,\dots i_1) = M_{b+1,i_k}M_{i_k,i_{k-1}}\times\dots M_{i_2,i_1}\times N_{i_1,a-1}.$$

We write

$$M_{b+1,a-1}\coloneqq \bigcup_{b\le p\le a} M_{b+1,p}\times N_{p,a-1}\coloneqq \operatorname{colim} D.$$

We define composition category as same as the framed case.

resolve (wha... is the comp. cat. ?)

Theorem A. *Let $\mathscr{C}$ be a flow category with an $\mathfrak{X}$-tangential structure. There is a spectral sequence:*

$$E_1^{n,p} = \bigoplus_{|z|=p} \Omega^{\mathfrak{X}}_{n-p}$$

which conditionally converges to the homotopy groups of the realization:

resolve: by Lurie we ahve that the S.S con- vernce to th homotopy groups of

$$E_r^{n,p} \implies \pi_n |\mathscr{C}|\,.$$

An element

$$W_p = \{W_z\}_{|z|=p} \in E_1^{n,p}$$

survives to $E_{r+1}^{n,p}$ if and only if the collection W_p extends to a right flow module over $\mathscr{C}_{[p,p-r]}$ (see Definition 3.28). The differential is represented geometrically by the Toda manifolds:

move this ref after the thm?

$$d_{r+1}(W_p) = \bigcup_{y \in \mathscr{C}_{p-(r+1)}} (\mathcal{M}_{-,y} \circ W) \in \bigoplus_{y \in \mathscr{C}_{p-(r+1)}} \Omega^{\mathfrak{X}}_{n-p+r} = E_{r+1}^{n-1,p-(r+1)}$$

never justi- fied why thi colim is $|\mathscr{C}|$

Remark 7.7. The convergence of the spectral sequence in Theorem A relies on the convergence results established for chain complexes (Theorem B). Specifically, if the flow category $\mathscr{C}$ is bounded below, it induces a bounded below chain complex in $\mathrm{Ch}(\mathrm{Mod}_{\mathfrak{X}})$, which by Corollary 2.27 yields strong convergence. If we omit the bounded below condition on $\mathscr{C}$, we still obtain an exact couple that conditionally converges to the generalized homology of the realization, as detailed in Corollary 2.27.

7.1 Change of rings

Give a framed flow category $\mathscr{C}$, we can benefit from the fact that the changing the tangential structure of manifolds correspond to maps between spectra, to get a spectral sequence that calculates the bordism homology (represented by Thom spectra) of a manifold.

Let $\mathfrak{X}, \mathfrak{Y}$ be two tangential structures. Denote by $\mathbf{Flow}^{\mathfrak{X}}$ (resp. $\mathbf{Flow}^{\mathfrak{Y}}$) the collections of $\mathfrak{X}$ (resp. $\mathfrak{Y}$) structured flow categories.

A map $\mathfrak{X} \to \mathfrak{Y}$ induce a "forgetting" map F, from $\mathfrak{X}$ manifolds to $\mathfrak{Y}$ manifolds. The fact that Cartesian product commutes with F, induces a map:

$$(-)^{\mathfrak{Y}} : \mathbf{Flow}^{\mathfrak{X}} \to \mathbf{Flow}^{\mathfrak{Y}}$$

On the other hand, the map $\mathfrak{X} \to \mathfrak{Y}$ induces a map of spectra $M\mathfrak{X} \to M\mathfrak{Y}$ and hence a map of coherent chain complexes:

$$\otimes_{M\mathfrak{X}} M\mathfrak{Y} \colon\ \mathrm{Ch}(\mathrm{Mod}_{M\mathfrak{X}}) \to \mathrm{Ch}(\mathrm{Mod}_{M\mathfrak{Y}})$$

induces by post composition

$$\mathrm{Ch} \xrightarrow{K} \mathrm{Mod}_{M\mathfrak{X}} \xrightarrow{\otimes_{M\mathfrak{X}} M\mathfrak{Y}} \mathrm{Mod}_{M\mathfrak{Y}}$$

Lemma 7.8. *The following diagram commutes:*

Re-write the proof

$$\begin{tikzcd}
\mathbf{Flow}^{\mathfrak{X}} \arrow[r, "K_{(-)}"'] \arrow[d, "(-)^{\mathfrak{Y}}"'] & \mathrm{Ch}(Mod_{M\mathfrak{X}}) \arrow[d, "\otimes_{M\mathfrak{X}} M\mathfrak{Y}"] \\
\mathbf{Flow}^{\mathfrak{Y}} \arrow[r, "K_{(-)}"'] & \mathrm{Ch}(Mod_{M\mathfrak{Y}})
\end{tikzcd}$$

Proof. Let $\mathscr{C}$ be a $\mathfrak{X}$ flow category. On the one hand we have $K_{\mathscr{C}^{\mathfrak{Y}}}$, on the other $K_{\mathscr{C}} \otimes_{M\mathfrak{X}} M\mathfrak{Y}$. The object of the coherent chains are:

$$(K_{\mathscr{C}^{\mathfrak{Y}}})_p = \bigoplus_{|z|=p} M\mathfrak{Y} = \bigoplus_{|z|=p} M\mathfrak{X} \otimes_{M\mathfrak{X}} M\mathfrak{Y} \simeq (K_{\mathscr{C}} \otimes_{M\mathfrak{X}} M\mathfrak{Y})_p.$$

We have that the maps of $K_{\mathscr{C}^{\mathfrak{Y}}}$ are the collapse maps:

$$c_{M^{\mathfrak{Y}}_{z,w}} : \Sigma^{\infty}_{+}\mathbb{R}^{J_{z,w}}_{0\leq} \to M\mathfrak{Y} \tag{38}$$

where $M^{\mathfrak{Y}}$ is the manifold M with $\mathfrak{Y}$ structure. The maps of $K_{\mathscr{C}} \otimes_{M\mathfrak{X}} M\mathfrak{Y}$ are given by collapsing and tensoring

$$c_{M_{z,w}} \otimes_{M\mathfrak{X}} M\mathfrak{Y} : \Sigma^{\infty}_{+}\mathbb{R}^{J_{z,w}}_{0\leq} \to M\mathfrak{X} \xrightarrow{(-)\otimes_{M\mathfrak{X}} M\mathfrak{Y}} M\mathfrak{Y}. \tag{39}$$

The fact that the $\mathfrak{Y}$ structure of $M_{z,w}$ is coming from $\mathfrak{X}$ yields that the collapse map of $M^{\mathfrak{Y}}_{z,w}$ factors through $\mathfrak{X}$, the functoriality of the collapse map give the equivalence between Eq. (38) and Eq. (39). ☐

First, we get that we can start with framed flow category and forget to other tangential structure, to get that:

Corollary 7.9. *Let $\mathscr{C}$ be a framed flow category and let $\mathfrak{X}$ be a tangential structure. Then the spectral sequence of Theorem A implies to $\mathscr{C}^{\mathfrak{X}}$ having:*

$$E_r \implies \pi_*(|\mathscr{C}^{\mathfrak{X}}|) \simeq \pi_*(|\mathscr{C}| \otimes M\mathfrak{X}) =: M\mathfrak{X}_*(|\mathscr{C}|)$$

The first equivalence holds because the realization from Ch to $\widehat{\mathrm{Fil}}$ is given by colimit, hence commutes with tensoring (as left adjoint).

Second, we can apply it to Morse function, to calculate the bordism homology of a manifold:

Corollary 7.10. *Let $f : M \to \mathbb{R}$ be a Smale Morse function and let $\mathfrak{X}$ be a tangential structure, then the spectral sequence of the framed flow category $(\mathscr{C}_f)^{\mathfrak{X}}$ converge to $M\mathfrak{X}_*(M)$.*

Remark 7.11. Since the Morse index is inherently bounded below (by 0), the induced flow category $\mathscr{C}_f$ is bounded below. Thus, Corollary 2.27 guarantees strong convergence to the bordism homology.

7.2 Bordism homology of projective spaces

Let $A = MO$, the unoriented bordism spectrum. We remind that

$$MO_* = \mathbb{Z}/2\left[x_i;\ i \neq 2^k - 1\right]$$

Let $a \in \mathbb{R}^{n+1}$ be a weights vector such that $a_i < a_{i+1}$ and let $\bar{f} : \mathbb{R}^{n+1} \to \mathbb{R}$ be the weighted distance function:

$$\bar{f}(x) = \sum a_i x_i^2$$

Let $M = \mathbb{RP}^\ell$,notice that $\bar{f}(x) = \bar{f}(\lambda x)$, therefore $\bar{f}$ induce a function $f : M \to \mathbb{R}$.

Notice that the critical points of f are given by:

$$e_i \coloneqq [0 : \cdots : 0 : \underbrace{1}_{i} : 0 : \cdots : 0]$$

with index:

$$|e_i| = i - 1.$$

Denote by $E_r^{n,p}$ the spectral sequence of Corollary 7.10 with coefficients in MO. We get that E_1 is given by:

$$E_1^{n,p} = \Omega_{n-p}^O$$

To calculate the flow manifolds, first notice that the stable manifolds are given by:

$$W_p^s = [0 : \cdots : 0 : \underset{\neq 0}{x_p} : x_{p+1} : \cdots : x_\ell] \tag{40}$$

and the unstable manifolds are given by:

$$W_p^u = [x_0 : \cdots : x_{p-1} : \underset{\neq 0}{x_p} : 0 : \cdots : 0] \tag{41}$$

The intersection is:

$$W_q^u \cap W_p^s = [0 : \cdots : 0 : \underset{\neq 0}{x_p} : x_{p+1} : \cdots : x_{q-1} : \underset{\neq 0}{x_q} : 0 : \cdots : 0] \tag{42}$$

By solving the flow equation we get that the action of the flow is given by:

$$\phi_t([x_i]) = [e^i t x_i]$$

So we get that the flow acts on $W_q^u \cap W_p^s$ by:

$$\begin{aligned} &[0 : \cdots : 0 : \underset{\neq 0}{x_p} : x_{p+1} : \cdots : x_{q-1} : \underset{\neq 0}{x_q} : 0 : \cdots : 0] \mapsto \\ &[0 : \cdots : 0 : \underset{\neq 0}{x_p} : e^t x_{p+1} : \cdots : e^{(q-p-1)t} x_{q-1} : \underset{\neq 0}{e^{(q-p)t} x_q} : 0 : \cdots : 0] \end{aligned}$$

By using the scalar multiplication in $\mathbb{RP}^\ell$ we can assume that $x_p = 1$. Moreover, by taking quotient by the flow ϕ_t we can assume that in the quotient:

$$W_q^u \cap W_p^s / \phi_t$$

the q coordinate is ± 1. We conclude that the (non compactified) flow manifolds are:

$$M'_{q,p} = \{[0 : \cdots : 0 : 1 : x_{p+1} : \cdots : x_{q-1} : \pm 1 : 0 : \cdots : 0]\}$$

And we get that after Compactification

Proposition 7.12. *The flow manifold of $\mathbb{RP}^\ell$ with the Morse function $\bar{f}(x) = \sum a_i x_i^2$ are given by two cubes:*

$$M_{q,p} = S^0 \times I^{q-p-1}$$

We get that the first differentials are multiplication by

$$M_{p,p-1} = I^{q-p-1} \sqcup I^{q-p-1} = S^0 \sim \varnothing \in \Omega_0^O.$$

So the first differential is zero, hence every element $X \in \Omega_{n-p}^O$ survives. We can also see this by Corollary 7.10, as we can always extend in the following way:

$$\begin{tikzcd} \bullet \arrow[r, "X"] \arrow[rr, bend right, "X\times I"'] & \bullet \arrow[r, "S^0"] & \bullet \end{tikzcd}$$

To calculate the second differential we need to calculate the Toda brackets of the two flow categories:

$$\begin{tikzcd} \bullet \arrow[r, "X"] \arrow[rr, bend right, "X\times I"'] & \bullet \arrow[r, "S^0"] & \bullet \end{tikzcd} \qquad \begin{tikzcd} \bullet \arrow[r, "S^0"] \arrow[rr, bend right, "S^0\times I"'] & \bullet \arrow[r, "S^0"] & \bullet \end{tikzcd}$$

Which is given by the colimit of the diagram:

$$\begin{tikzcd} & X \times S^0 \times S^0 \arrow[dl] \arrow[dr] & \\ X \times I \times S^0 & & X \times S^0 \times I \end{tikzcd}$$

we get the topological space $X \times \partial I^2$, which after straightening the corners (Lemma 6.3) is the manifold $X \times S^1$.

Proposition 7.13. *Let E be the spectral sequence of Let $\mathbb{RP}^\ell$ with the morse function $\bar{f}(x) = \sum a_i x_i^2$ and let MO be the trivial tangential structure and let $E_r^{n,p}$ be the spectral sequence arising from Theorem A. Then $E_r^{n,p}$ collapse at the first page and we have that*

$$MO_i(\Sigma_+^\infty \mathbb{RP}^\ell) = \bigoplus_{\ell-i \le j \le i} MO_j.$$

Proof. We need to show that for any $r \in \mathbb{N}$ and for any $X \in E_1$, we have that X survives to E_r.

Denote by $\mathscr{C}$ the flow category of f. By Corollary 7.10, we need to extend X to a right flow module W of degree n over $\mathscr{C}_{[p,p-r]}$. We define the manifolds of W to be:

$$W_q = X \times I^{p-q}.$$

The equations

$$\partial W_q = X \times \partial I^{p-q} = \bigcup X \times I^{p-q-1} \times S^0 = \bigcup_{q<q' \le p} X \times I^{p-q'} \times S^0 \times I_{q'-q-1} = \bigcup_{q<q' \le p} W_{q'} \times M_{q',q}$$

justify that W is indeed a right flow module over $\mathscr{C}_{[p,p-r]}$.

Moreover, the differential is given by the Toda manifold $W \circ \mathcal{M}_{-,p-r-1}$, which is:

$$d_{r+1}(X) = \bigcup_{p \geq q \geq p-r} W_q \times M_{q,p-r-1} = \bigcup_q X \times I^{p-q} \times S^0 \times I^{q-(p-r-1)-1} = \bigcup_q X \times I^r \times S^0 = X \times \partial I^{r+1} \simeq X \times S^r \tag{43}$$

□

Now we want to calculate $MSO(\Sigma_+^\infty \mathbb{RP}^\ell)$. Under the same settings we have

Proposition 7.14. *Let E be the spectral sequence of $\mathbb{RP}^\ell$ with the morse function $\bar{f}(x) = \sum a_i x_i^2$ and coefficient ring MSO. We get that*

$$E_1^{n,p} = \Omega_{n-p}^{SO}$$

The first differential is given by multiplication with

$$d_2^{n,p} = \begin{cases} +,+ & p \text{ is even} \\ +,- & p \text{ is odd} \end{cases}.$$

Moreover every element $X \in E_2$ survives to E_∞ and the reduce homology is:

$$\Omega_i^{SO}(\mathbb{RP}^\ell) = \begin{cases} \Omega_{i-\ell}^{SO}[2] \oplus \Omega_{i-\ell+1}^{SO}/2 \oplus \Omega_{i-\ell+2}^{SO}[2] \cdots \oplus \Omega_{i-1}^{SO}/2 & \ell = 2k \\ \Omega_{i-\ell}^{SO} \oplus \Omega_{i-\ell+1}^{SO}[2] \oplus \Omega_{i-\ell+2}^{SO}/2 \oplus \Omega_{i-\ell+3}^{SO}[2] \cdots \oplus \Omega_{i-1}^{SO}/2 & \ell = 2k+1 \end{cases}$$

where $\Omega_j^{SO}[2]$ are the elements $X \in \Omega_j^{SO}$ such that $2X$ is null bordant.

Proof. The existence of the spectral sequence following from Corollary 7.10. The framing, hence orientation, of the zero dimensional manifolds $M_{p,p-1}$ is induces from the exact sequence:

$$T(W_p^u \cap W_{p-1}^s) \to TW_p^u \to \nu(W_{p-1}^s)$$

We denote the two components of the zero dimensional flow manifold

$$M_{p,p-1}^+, M_{p,p-1}^- \in M_{p,p-1} = [0 : \cdots : 0 : 1 : \pm 1 : 0 \cdots : 0]$$

and we compare the framing of $M_{p,p-1}^+$ to the framing of $M_{p,p-1}^-$ by composing:

$$\varphi : T(W_p^u \cap W_{p-1}^s)_- \oplus \nu W_{p-1}^s \simeq TW_p^u \simeq T(W_p^u \cap W_{p-1}^s)_+ \oplus \nu W_{p-1}^s$$

Let $v_0, \ldots, v_{p-2}$ be a basis for νW_{p-1}^s (which is of size $p-1$), the maps

$$\nu(W_{p-1}^s) \to TW_p^u$$

are induces from the parallel transport

$$\nu_{p-1}(W_{p-1}^s) \to T_p W_p^u$$

(through the points $M_{p,p-1}^\pm$) hence $\varphi_{|_{\nu W_{p-1}^s}}$ is the holonomy along the curve

$$\gamma(t) = [0 : \cdots : 0 : \cos(t), \sin(t) : 0 : \cdots : 0], \ t \in [0, \pi]$$

By lifting γ to the sphere we get that:

$$\varphi(v_0, \dots, v_{p-1}) = -v_0, \dots, -v_{p-2}.$$

Combine that, with the definition:

$$M_{p,p-1} \coloneqq W_p^u \cap W_{p-1}^s / \nabla f$$

where the flow ∇f acts on $M_{p,p-1}^{\pm}$ in opposite directions, to conclude that for an element $X \in E_1^{n,p}$ we have that $d_1(X)$ is two copies of X where the framing on one copy is given by inverting p vectors comparing to the other copy. Hence

$$d_1^{n,p}(X) = \begin{cases} \bar{X} \sqcup X & p \text{ is odd} \\ X \sqcup X & p \text{ is even} \end{cases}$$

Let $X \in E_1^{n,p}$. For p odd we have that X extends to a right flow module W over $\mathscr{C}_{[p,p-1]}$ by setting:

$$W_{p-1} = X \times I.$$

As we defined the "forgetting" map $\mathbf{Flow}^{SO} \to \mathbf{Flow}^{O}$ induce that diagram:

$$\begin{tikzcd} (\mathcal{E}_r^{n,p})^{SO} \arrow[r, "d_{r+1}^{SO}"] \arrow[d, "F"'] & (\mathcal{E}_r^{n-1,p-r-1})^{SO} \arrow[d, "F"] \\ (\mathcal{E}_r^{n,p})^{O} \arrow[r, "d_{r+1}^{O}"'] & (\mathcal{E}_r^{n-1,p-r-1})^{O} \end{tikzcd}$$

where F is the forgetting map $\Omega^{SO} \to \Omega^{O}$ and $\mathcal{E}$ are the elements that survive to the $(r+1)$-th page in the (un)oriented spectral sequence. The flow module above shows that $X \in E_1^{n,p}$ survives to E_2, inductively, we assume that X survives to E_{r+1} (i.e., extends to a right module over $\mathscr{C}_{[p,p-r]}$). Using that diagram and Eq. (43), we get that

$$F \circ d_{r+1}^{SO}(X) \simeq d_{r+1}^{O} \circ F(X) \simeq F(X) \times S^r.$$

Both orientations of the circle are null-bordant, hence the underlying manifold of $d_{r+1}^{SO}(X)$ is null-bordant and X survives to E_{r+2} (i.e., the module extends over $\mathscr{C}_{[p,p-r-1]}$).

For p even, we get that $d_1(X) = X \sqcup X$, hence X survives to $E_2 \iff X$ is 2-torsion. Under this assumption, we can use the fact that d^{SO} lifts d^{O} to get that X survives to E_{r+1}.

We conclude that the spectral sequence collapse at the second page and

$$\Omega_i^{SO}(\mathbb{RP}^{\ell}) = \begin{cases} \Omega_{i-\ell}^{SO}[2] \oplus \Omega_{i-\ell+1}^{SO}/2 \oplus \Omega_{i-\ell+2}^{SO}[2] \cdots \oplus \Omega_{i-1}^{SO}/2 & \ell = 2k \\ \Omega_{i-\ell}^{SO} \oplus \Omega_{i-\ell+1}^{SO}[2] \oplus \Omega_{i-\ell+2}^{SO}/2 \oplus \Omega_{i-\ell+3}^{SO}[2] \cdots \oplus \Omega_{i-1}^{SO}/2 & \ell = 2k+1 \end{cases}$$

□

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